

INDUSTRIAL PROCESS MODELLING & SIMULATION TRAINING

2 – 4 August 2022

Part 4 : Reactor Modelling and Simulation, Flowsheet
Convergence and Property Sets

About Myself

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Topics

4.1 Reactors

- Types of Reactor Model in Aspen Plus
- Simulation of Balanced Based Reactor Model
- Simulation of Kinetic Based Reactor Model
- Achieving Process Objectives using Design Specifications

4.2 Flowsheet Convergence

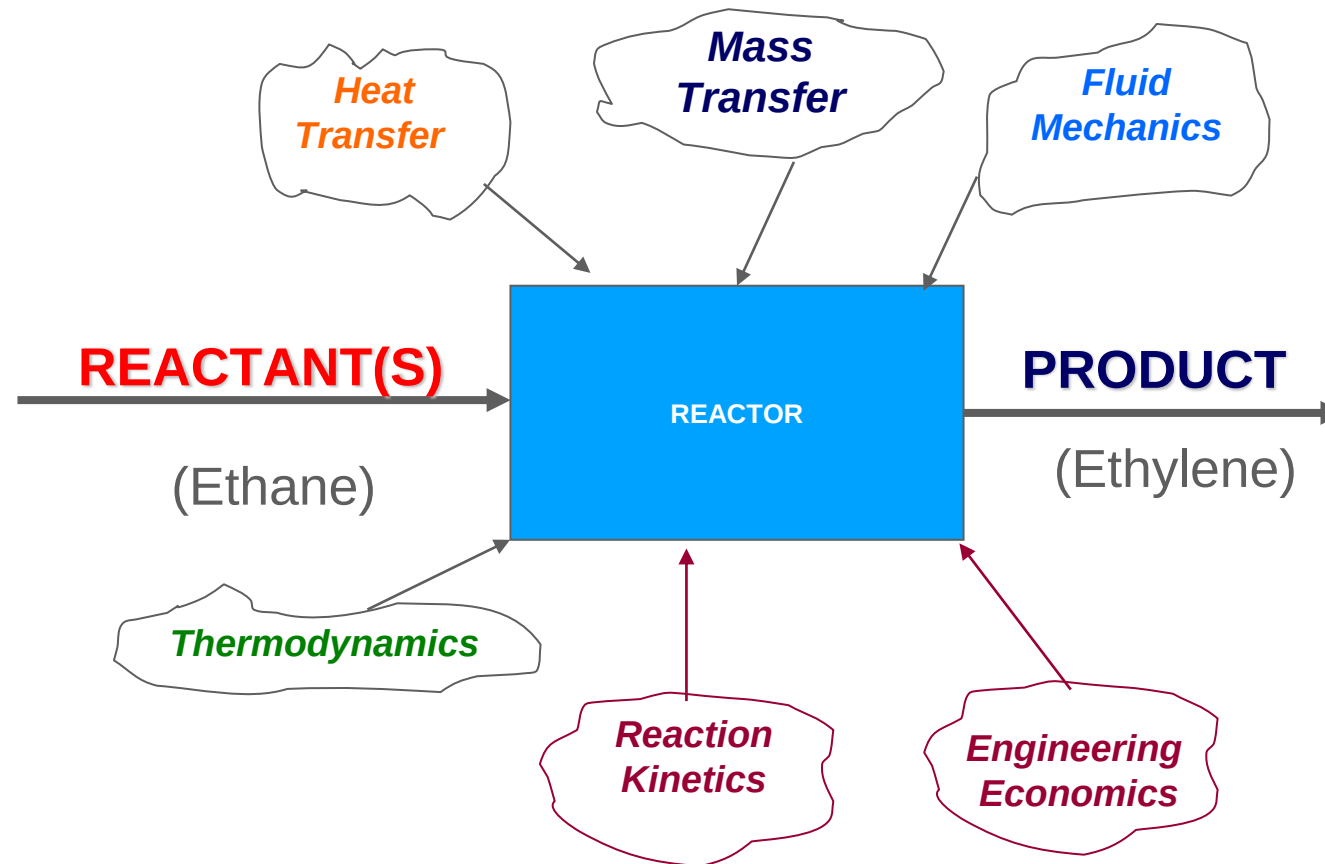
4.3 Property Sets

Outcomes

- Able to simulate balance based reactor and kinetic based reactors and solve problems related to reaction process
- Able to apply design specification function in achieving the target specification
- Able to identify the non-convergence problem
- Able to use Property Sets to access properties

Introduction

What's involved in reactor design ?



Basic of Reaction Performances

$$\text{Conversion, } X \text{ (fraction basis)} = \frac{F_{A0} - F_A}{F_{A0}}$$

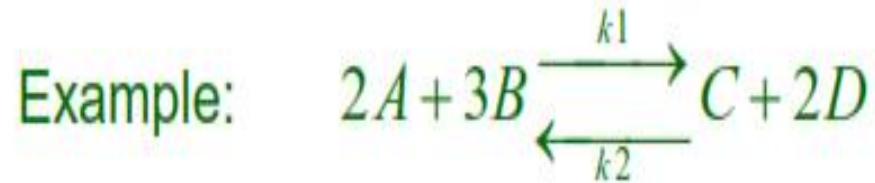
$$\text{Selectivity, } S = \frac{F_P}{F_{A0} - F_A}$$

$$\text{Yield, } Y = \frac{F_P}{F_{A0}}$$

where F_A = flowrate of reactant A, F_{A0} = initial flowrate of reactant A,
 F_P = flowrate of product P

Basic of Reaction Kinetics

Delineating the 1. reaction mechanism and 2. rate of reaction



Forward reaction: (Assuming the reaction is 2nd order in A)

coefficients: A: -2 B: -3 C: 1 D: 2

exponents: A: 2 B: 0 C: 0 D: 0

Stoichiometric
coefficient

Order of reaction =
second order in C_A
 $r_f = kC_A^2$

Reverse reaction: (Assuming the reaction is 1st order in C and D)

coefficients: C: -1 D: -2 A: 2 B: 3

exponents: C: 1 D: 1 A: 0 B: 0

Power-Law Rate Expression

$$rate = k \times \prod_i [concentration_i]^{exponent_i}$$

$$k = (\text{Pre-exponential Factor}) \left(\frac{T}{T_0} \right)^n \exp \left(- \frac{\text{Activation Energy}}{R} \left[\frac{1}{T} - \frac{1}{T_0} \right] \right)$$

If reference temperature, T_0 is not specified, k is expressed as:

$$k = (\text{Pre-exponential Factor}) T^n \exp \left(- \frac{\text{Activation Energy}}{RT} \right)$$

When $n=0$, then

$$k = A \exp \left(- \frac{E}{RT} \right)$$

where A =pre-exponential factor, E = activation energy

When $n=0$, then

$$k = A \exp\left(-\frac{E}{RT}\right)$$

Where A=pre-exponential factor, E= activation energy



Format of the kinetic information in Aspen Plus

Kinetic factor

If T_0 is specified Kinetic factor $= k(T/T_0)^n e^{-(E/R)[1/T-1/T_0]}$

If T_0 is not specified Kinetic factor $= kT^n e^{-E/RT}$

Meaning,

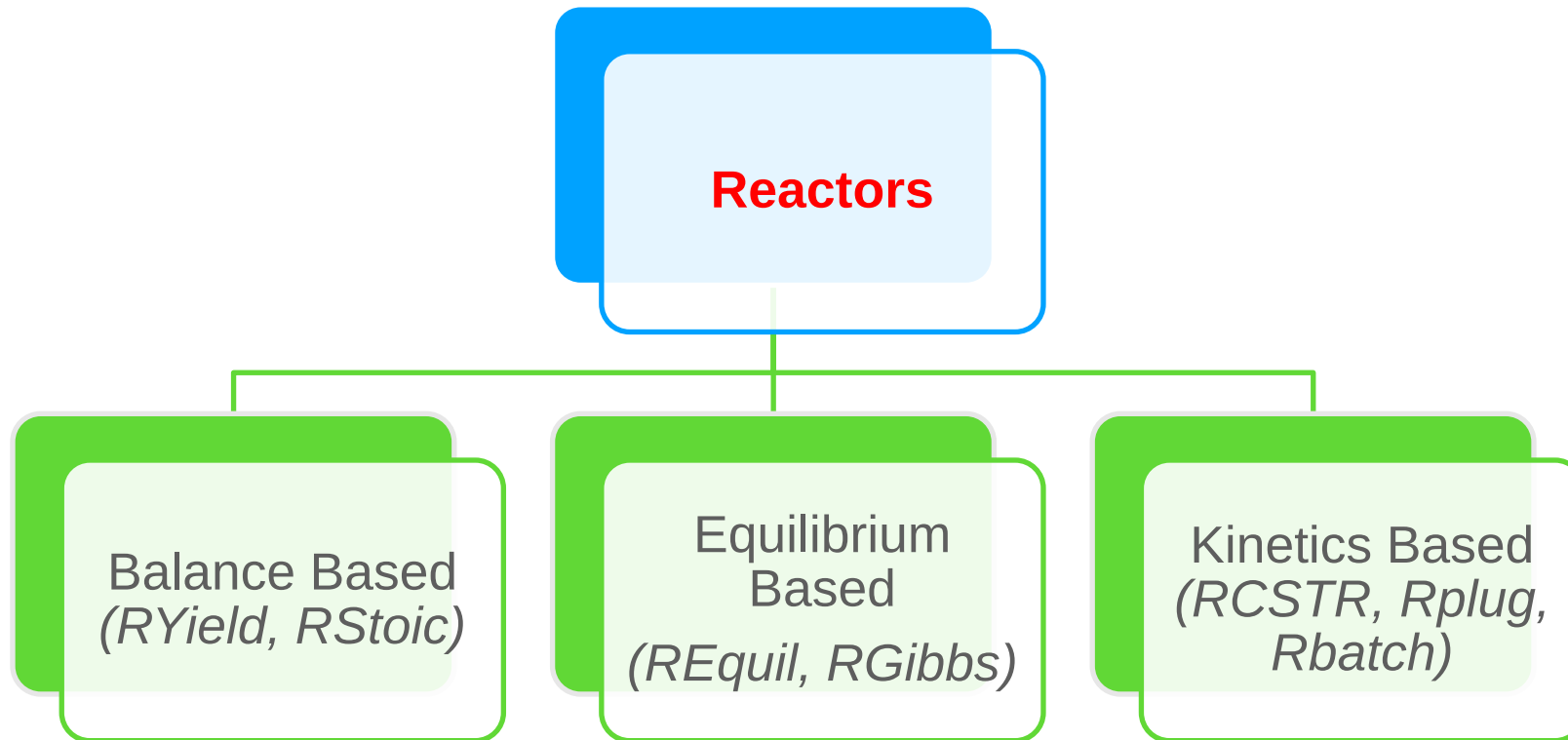
Kinetic factor = k

$k = A$, the pre-exponential factor

4.1 Reactors

Reactor Overview

12



RPLUG, RCSTR and RBATCH **are rigorous models** for plug flow, CSTR and batch reactors, respectively.

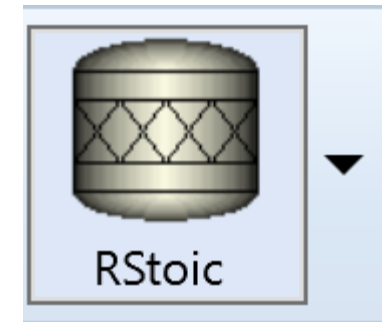
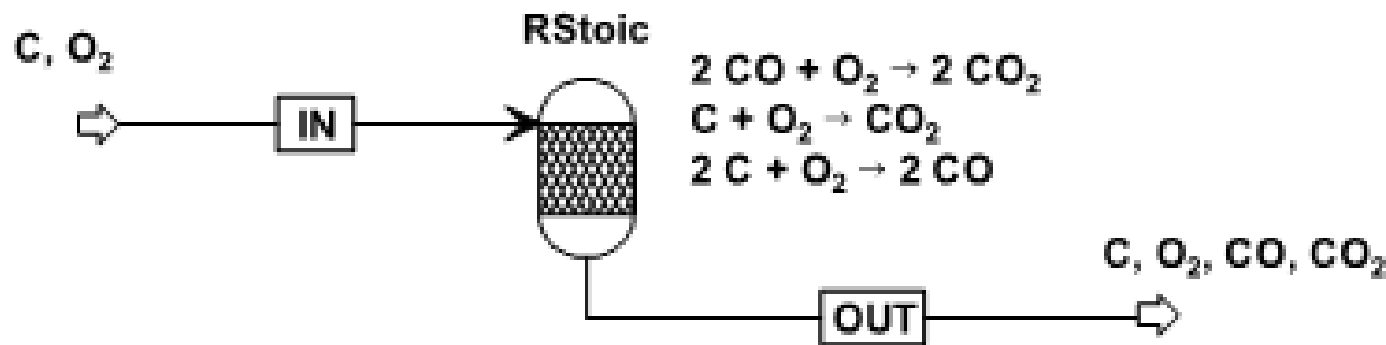
These reactor models are the commonly **used to model reaction chemistry**.

Reactor Block

Model	Description	Purpose	Use
RStoic	Stoichiometric reactor	Stoichiometric reactor with specified reaction extent or conversion	Reactors where the kinetics are unknown or unimportant but stoichiometry and extent are known
RYield	Yield reactor	Reactor with specified yield	Reactors where the stoichiometry and kinetics are unknown or unimportant but yield distribution is known
REquil	Equilibrium reactor	Chemical and phase equilibrium by stoichiometric calculations	Single- and two-phase chemical equilibrium and simultaneous phase equilibrium
RGibbs	Equilibrium reactor	Chemical and phase equilibrium by Gibbs energy minimization	Chemical and/or simultaneous phase and chemical equilibrium. Includes solid phase equilibrium.
RCSTR	Continuous stirred tank reactor	Continuous stirred tank reactor	One, two, or three-phase stirred tank reactors with kinetics reactions in the vapor or liquid
RPlug	Plug flow reactor	Plug flow reactor	One, two, or three-phase plug flow reactors with kinetic reactions in any phase. Plug flow reactions with external coolant.
RBatch	Batch reactor	Batch or semi-batch reactor	Batch and semi-batch reactors where the reaction kinetics are known

Balanced Based reactors - *Rstoic*

- Requires both an atom and a mass balance
- Used in situations where both the equilibrium data and the kinetics are either unknown or unimportant
- RStoic can model reactions occurring simultaneously or sequentially.



Balanced Based reactors - *Rstoic*

- Must specify the stoichiometry for each reaction and
 - Molar extent of reaction
 - Fractional conversion based on a reactant
- Combustion option to generate combustion reactions
 - Assumes full combustion of carbon, hydrogen and nitrogen in feed
 - Oxygen must be specified in the Components list
 - Components containing atoms other than carbon, hydrogen, nitrogen, oxygen and sulfur are ignored
 - Option for NO or NO₂ for nitrogen combustion product

Balanced Based reactors - *Rstoic*

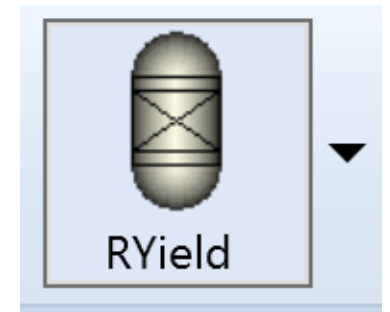
- Can specify or report calculated heat of reaction at a reference temperature, pressure and phase
 - Why does the Reference Phase matter?

Heats of Reaction

- Heat of reaction need not be provided for reactions
- If you have a heat of reaction value that does not match the value calculated by Aspen Plus, you can adjust the heats of formation (**DHFORM**) of one or more components to make the heats of reaction match
- Heats of reaction can also be calculated or specified at a reference temperature and pressure in an RStoic reactor

Balanced Based Reactors - *RYield*

- Requires a mass balance only, not an atom balance
- Turn off atom balance check on **Block Options | Simulation Options** sheet
- Used to simulate reactors in which inlets to the reactor are not completely known but outlets are known
- Will normalize yields if necessary to maintain mass balance
- Specify product yields on a per mass of total feed (excluding inerts)
 - Option user Fortran subroutine to calculate yields

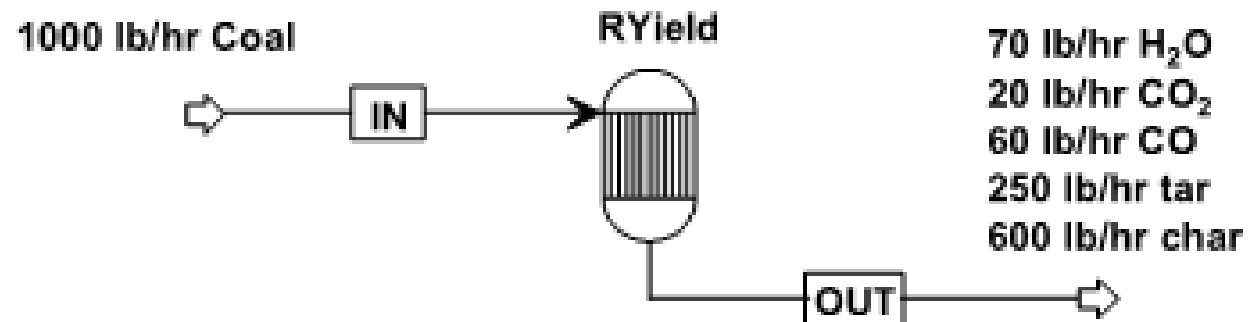


Balanced Based Reactors - *RYield*

- Component mapping to lump or de-lump components to expand or reduce the Component list
 - Can be used in EO modelling to control problem size and robustness
- For a lumping reaction, specify a list of reactants which are lumped to form the selected product
- For de-lumping reaction, the selected reactant delumps into a specified list of product

Balanced Based Reactors - *RYield*

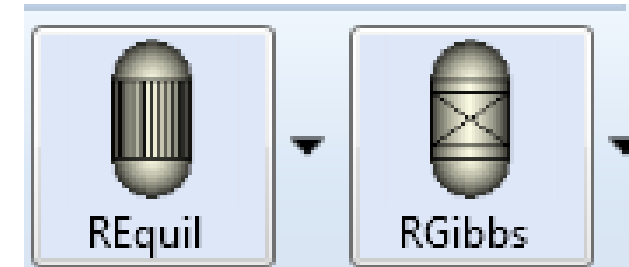
- Option to specify solid component attributes and PSD in product stream
- Used often to react non-conventional components like coal and wood pulp



Equilibrium Based Reactors

Requil and RGibbs

- Do not take reaction kinetics into account
- Solve similar problem, but specifications are different
- Allow individual reactions to be at a restricted equilibrium



Equilibrium Based Reactors - *REquil*

- Computes combined chemical and phase equilibrium by solving equilibrium equations
 - K_{eq} is calculated from Gibbs free energy: $\Delta G = - RT \ln K_{eq}$
- Useful when there are many components, a few known reactions, and when relatively few components take part in the reactions
- Cannot do a three-phase flash
- Specify the reaction stoichiometry along with the reactor conditions

Equilibrium Based Reactors - *RGibbs*

- Useful when reactions occurring are not known or are high in number, due to many components participating in the reactions
- A Gibbs free energy minimization is done to determine the product composition, at which the Gibbs free energy of the products is at a minimum
- With restricted equilibrium, a full set of linearly independent reactions is needed
- *RGibbs* can also calculate the chemical equilibria between any number of conventional solid components and the fluid phases

Equilibrium Based Reactors - *RGibbs*

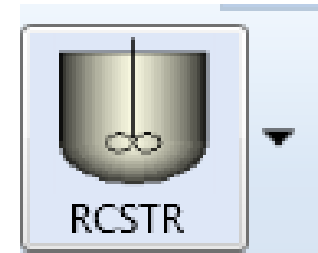
- Can use the **Setup | Products** sheet to assign different sets of species to each solution phase
- Option to assign phases to specific output streams on the **Setup | Assign Streams** sheet
- Select the appropriate Calculation option
- Specify pressure and temperature or duty
- Select the appropriate number of fluid phases (2 for V/L)

Kinetic Reactors

- Reaction kinetics describe the rate of disappearance of reactant and the formation of product species
- Kinetic (rate controlled) reactors are RCSTR, Rbatch, Rplug, Fluidbed and BatchOP
- Kinetics can be specified using one of the following built-in models:
 - ✓ Power Law
 - ✓ Langmuir-Hinshelwood-Hougen-Watson (LHHW)
 - ✓ General (Equilibrium, Power Law, LHHW and Custom)
 - ✓ Polymerization kinetics (step-growth, free-radical, Ziegler-Natta, etc.)
 - ✓ User defined model using Fortran or Aspen Custom Modeler
- A catalyst for a reaction can have a stoichiometric coefficient of zero
- Reactions are specified by referencing a global reaction model

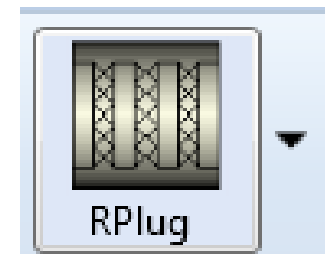
Kinetic Based Reactor – *RCSTR*

- Used when reaction kinetics are known and when the reactor contents have same properties as outlet stream
- Can model equilibrium reactions simultaneously with rate-based reactions
- Enter pressure and either temperature, duty or vapor fraction
- Reaction phase and hold-up must be specified
 - Volume
 - Residence time
 - Volume fraction



Kinetic Based Reactor – *RPlug*

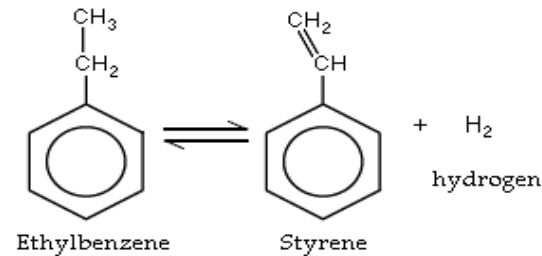
- Handles only rate-based reactions, including reactions with solids
- Assume that perfect mixing occurs in the radial direction and that no mixing occurs in the axial direction
- Option to use a user subroutine for kinetics, heat transfer, pressure drop and holdup
- A cooling stream is allowed
- Specify reactor type and reactor length and diameter
 - Number of tubes for multiple tube option



Simulation of Balanced Based Reactor Model

Example 4.1

Styrene is made from the dehydrogenation of ethylbenzene:



The feed stream to the reactor having a composition of pure ethylbenzene at 217gmol/s, 880 K, 1.378 bar. The process occur in isothermal reactor with pressure and temperature of 1.378 bar and 250°C respectively. Create a stoichiometric reactor model to be run isothermally. Use Peng-Rob property package.

Isothermal reactor: Fractional conversion of ethylbenzene, 0.8.

Problem Analysis

- Hydrocarbon system: Peng Rob property method is chosen
- Fractional conversion of ethylbenzene, 0.8: R-Stoic reactor model is chosen

Simulation steps

1. Add the chemical components
2. Select the property method (thermodynamics model)
3. Add a reactor unit (RStoic)
4. Specify the feed input
5. Include the reactor block specification, reaction details and heat of reaction
6. Run the simulation and check the results
7. Save the file in .bkp format

- Create blank simulation> include components>select property method> generate property and save the file in bkp format

Step 1: Components - Specifications

Component ID	Type	Component name	Alias
EB	Conventional	ETHYLBENZENE	C8H10-4
STY	Conventional	STYRENE	C8H8
H2	Conventional	HYDROGEN	H2

Step 2: Methods - Specifications

Property methods & options

- Method filter: COMMON
- Base method: PENG-ROB

Petroleum calculation options

- Free-water method: STEAM-TA
- Water solubility: 3

Electrolyte calculation options

- Chemistry ID:
- ☒ Use true components

Method name: PENG-ROB

EOS: ESPRSTD

Data set: 1

Liquid gamma:

Data set:

Liquid molar enthalpy: HLMX106

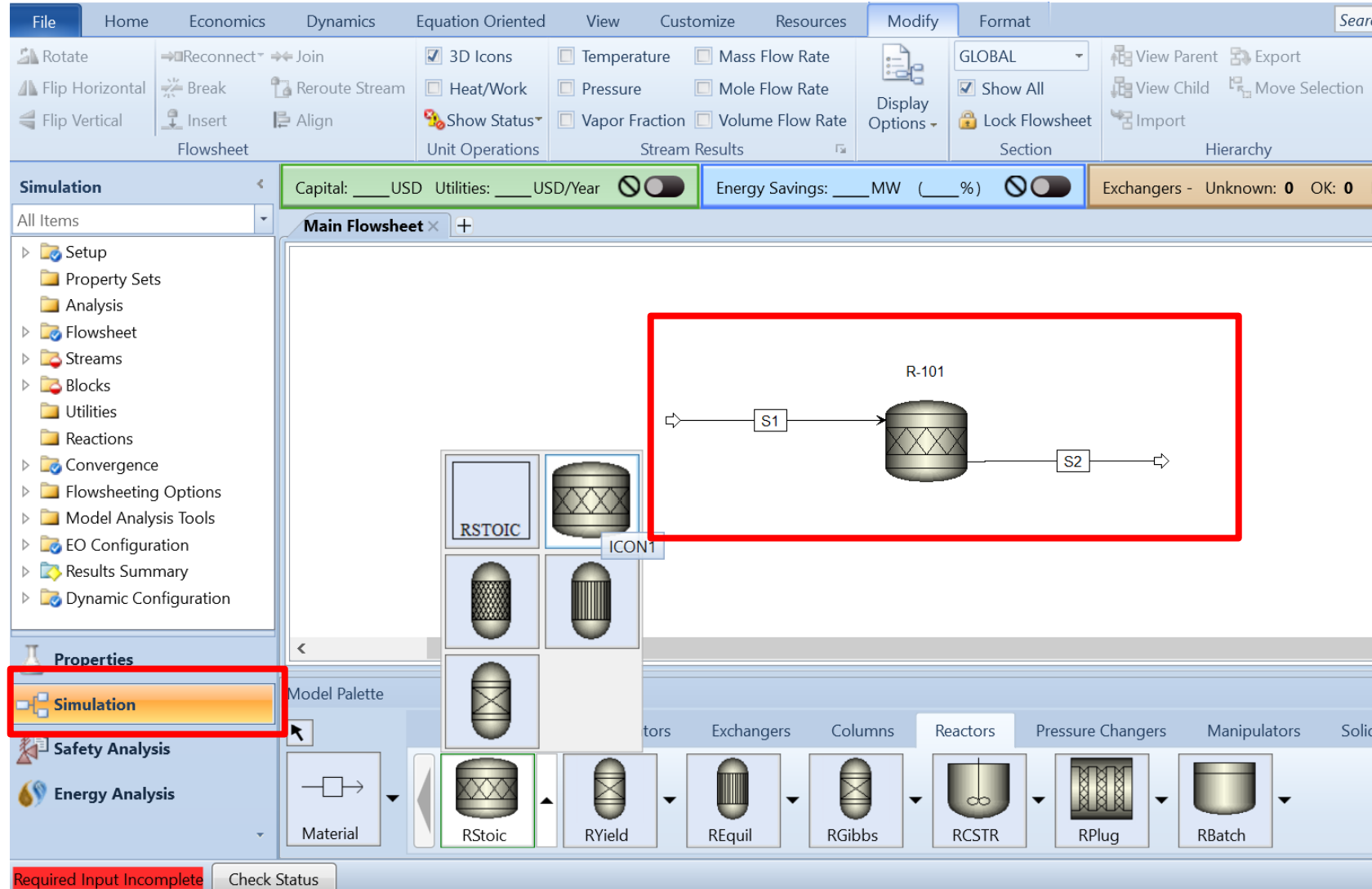
Liquid molar volume: VLMX20

☐ Heat of mixing

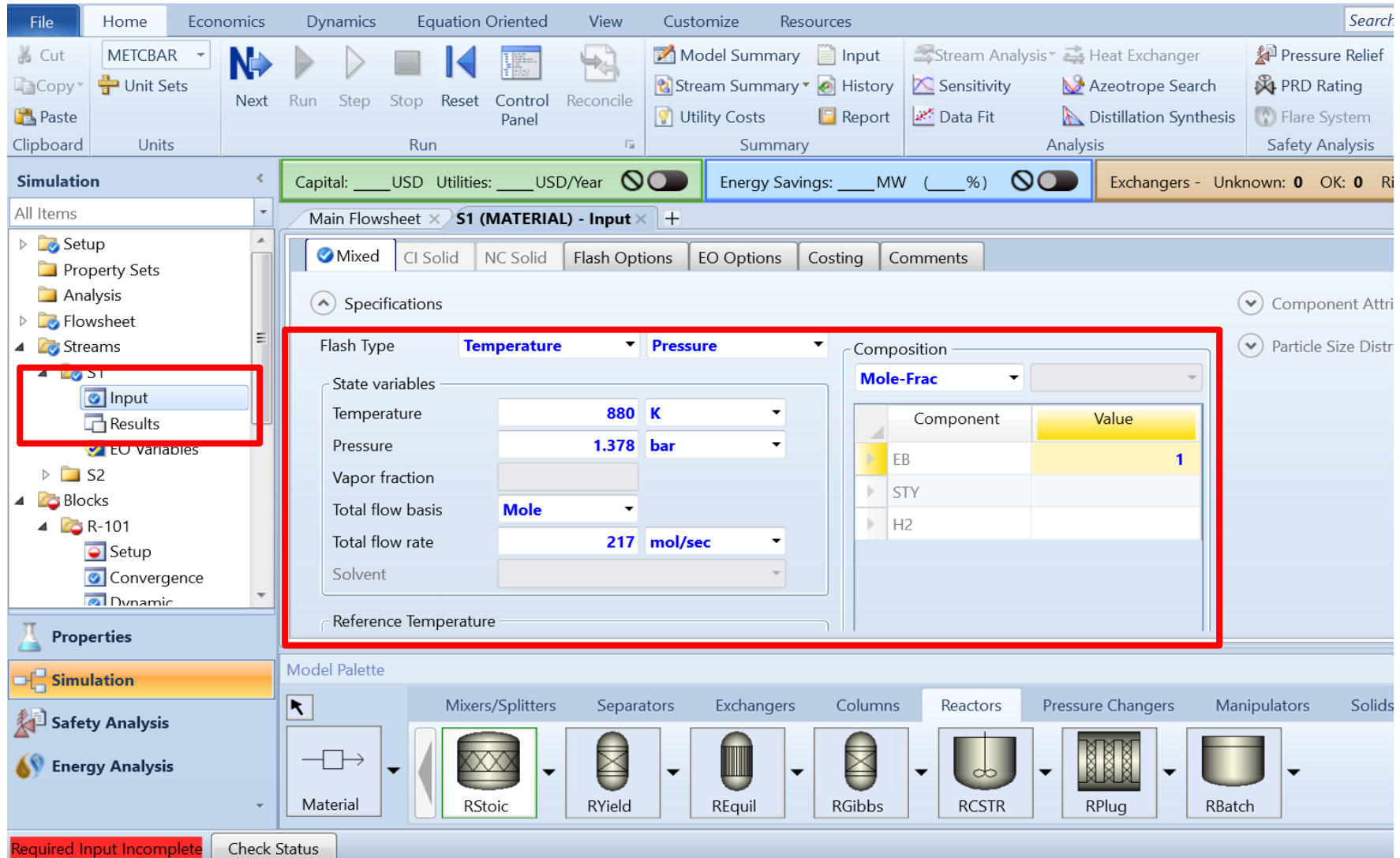
☐ Poynting correction

☐ Use liquid reference state enthalpy

- Shift to simulation environment > **construct the flowsheet**



- Shift to simulation environment> construct the flowsheet> **include the feed specification**



Capital: ___USD Utilities: ___USD/Year Energy Savings: ___MW (___%) Exchangers - Unknown: 0 OK: 0 Ri

Main Flowsheet x S1 (MATERIAL) - Input x

Mixed CI Solid NC Solid Flash Options EO Options Costing Comments

Specifications

Flash Type Temperature Pressure

State variables

Temperature 880 K

Pressure 1.378 bar

Vapor fraction

Total flow basis Mole

Total flow rate 217 mol/sec

Solvent

Reference Temperature

Composition

Mole-Frac

Component	Value
EB	1
STY	
H2	

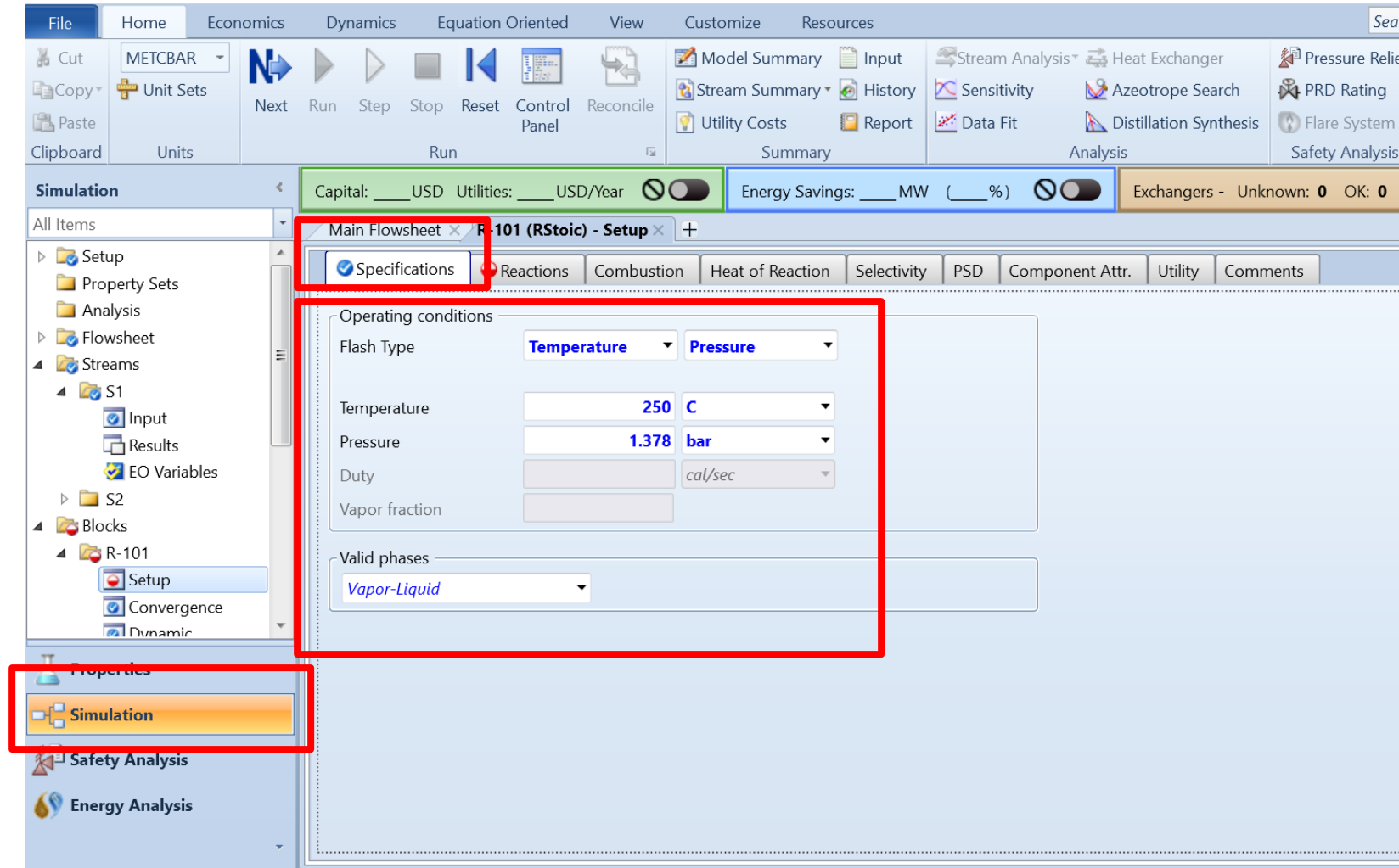
Model Palette

Mixers/Splitters Separators Exchangers Columns Reactors Pressure Changers Manipulators Solids

Material RStoic RYield REquil RGibbs RCSTR RPlug RBatch

Required Input Incomplete Check Status

- Shift to simulation environment> construct the flowsheet> include the feed specification> **include the reactor block specification**



- Shift to simulation environment> construct the flowsheet> include the feed specification> include the reactor block specification> **include reaction details**

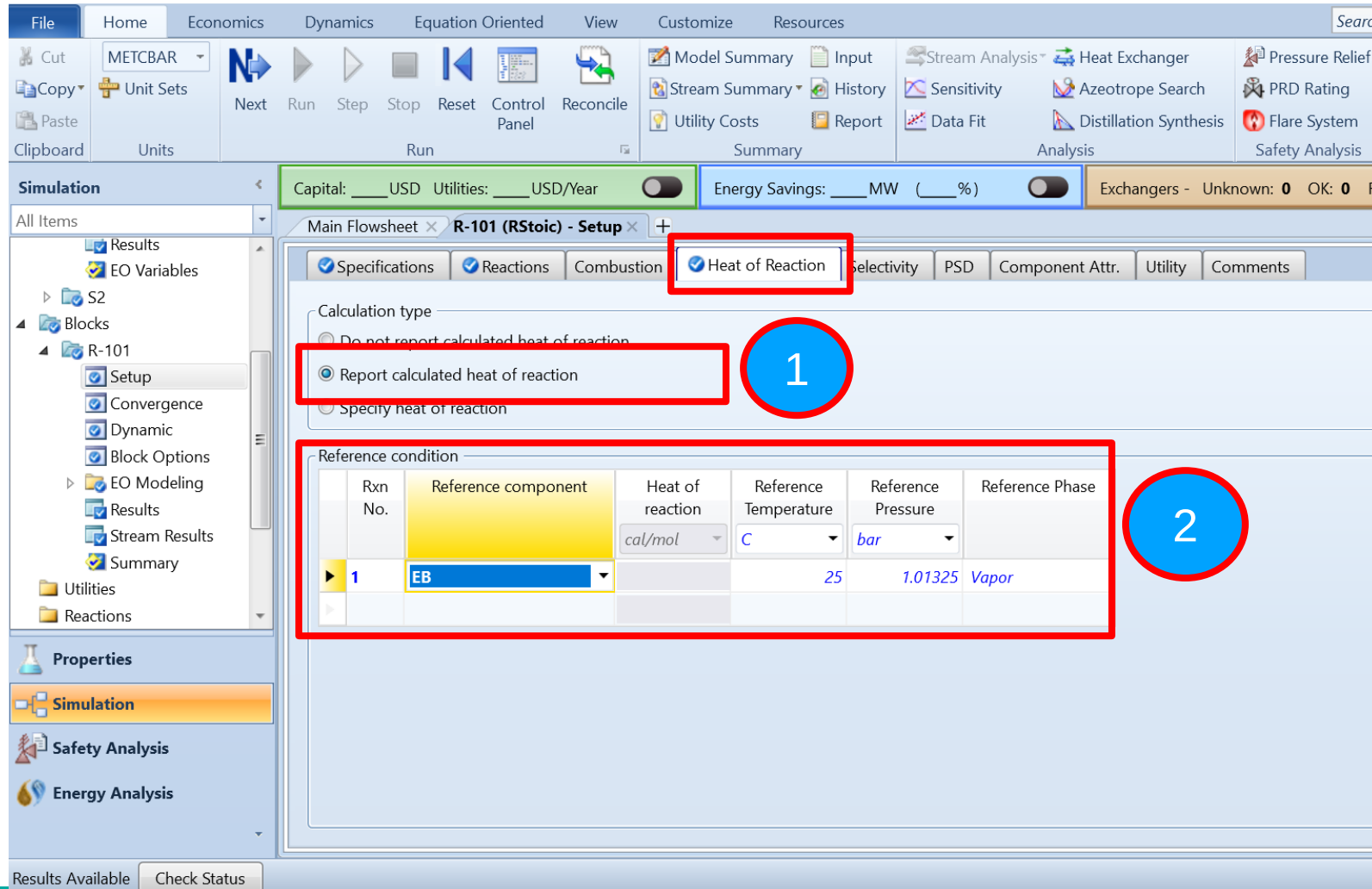
The screenshot displays the Aspen Plus software interface. The 'Edit Stoichiometry' dialog box is open, showing the configuration for reaction R-101. The 'Reactions' tab is selected, and the 'New' button is highlighted with a red circle and the number 1. The 'Reactants' table lists 'EB' with a coefficient of -1. The 'Products' table lists 'STY' and 'H2' with coefficients of 1. The 'Products generation' section shows 'Fractional conversion' selected with a value of 0.8 for component 'EB'. A red circle with the number 2 highlights the 'Products generation' section. The background shows the main flowsheet with various unit operations and a sidebar with a tree view of the process structure.

Component	Coefficient
EB	-1

Component	Coefficient
STY	1
H2	1

Products generation
Fractional conversion: 0.8 of component EB

- Shift to simulation environment> construct the flowsheet> include the feed specification> include the reactor block specification> include reaction details> **include the setting for heat of reaction**



Simulation

Capital: ___USD Utilities: ___USD/Year Energy Savings: ___MW (___%) Exchangers - Unknown: 0 OK: 0

Main Flowsheet R-101 (RStoic) - Setup

Specifications Reactions Combustion **Heat of Reaction** Selectivity PSD Component Attr. Utility Comments

Calculation type

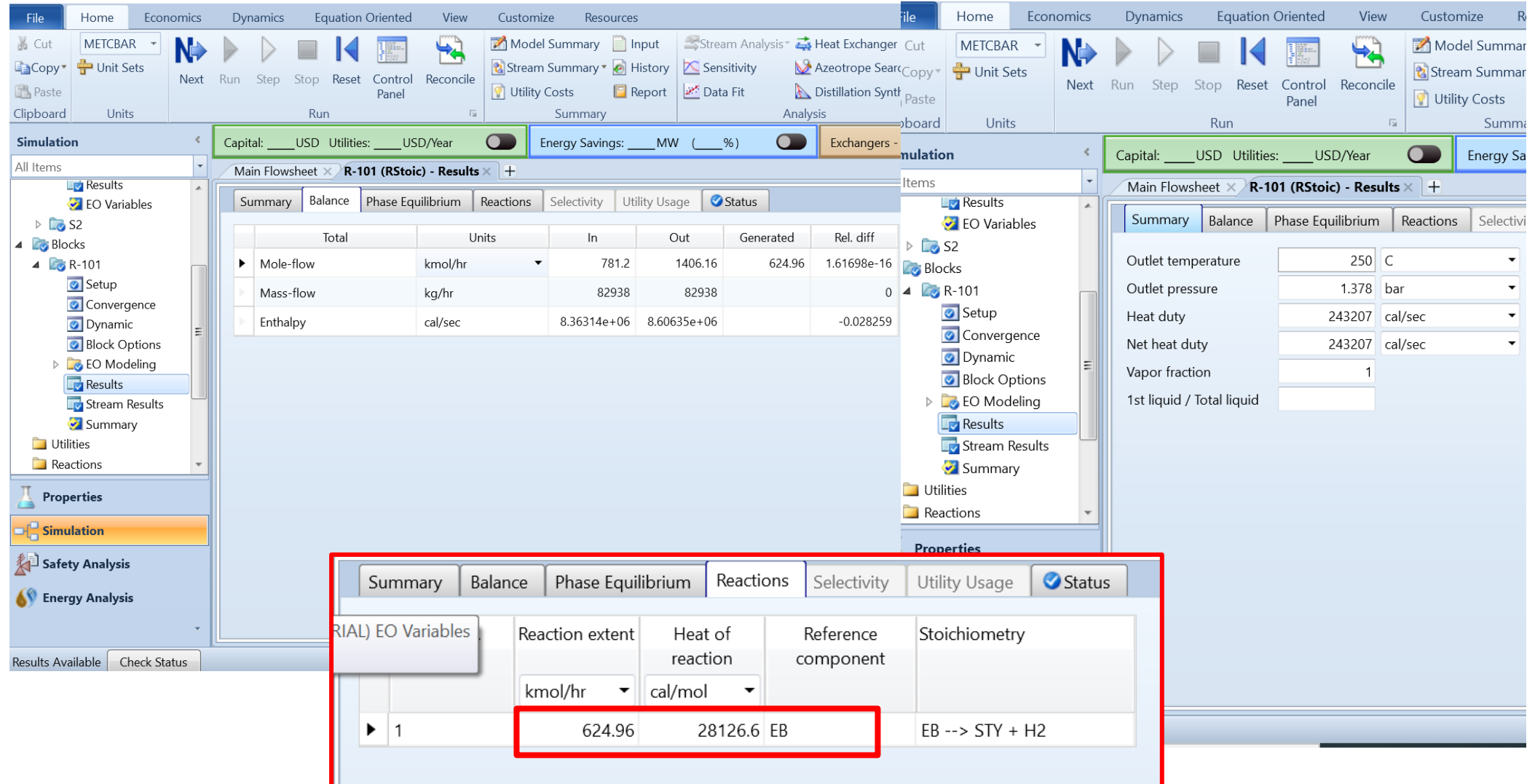
☐ Do not report calculated heat of reaction
☒ Report calculated heat of reaction
☐ Specify heat of reaction

Reference condition

Rxn No.	Reference component	Heat of reaction cal/mol	Reference Temperature C	Reference Pressure bar	Reference Phase
1	EB		25	1.01325	Vapor

Results Available Check Status

- Include the reactor block specification> include reaction details> include the setting for heat of reaction> **run simulation and check results**



The screenshot displays the Aspen Plus interface for a reactor simulation. The 'Reactions' tab is active, showing a table of reaction details. A red box highlights the reaction extent, heat of reaction, and reference component for reaction 1.

Reaction	Reaction extent	Heat of reaction	Reference component	Stoichiometry
1	624.96 kmol/hr	28126.6 cal/mol	EB	EB --> STY + H2

Simulation of Kinetic Based Reactor Model

Example 4.2

Crude glycerol can be converted to a value added product, 1-2 propanediol (PDIOL). The glycerol (GLY) is first dehydrated to acetol (ACL) and subsequently ACL is hydrogenated to PDIOL. The reactions involved are given as follows:



The rate for gas phase GLY dehydration, r_1 and ACL hydrogenation, r_2 are expressed through Eq. 3 and Eq. 4.

$$r_1 = k_1 P_{GLY} \quad \text{Eq. 3}$$

$$r_2 = k_2 P_{ACL} P_{H_2} \quad \text{Eq. 4}$$

where r is the rate of reaction in $\text{kmol}/(\text{m}^3.\text{s})$. P is the partial pressure in kPa for the respective component, k_1 and k_2 are the rate constant for the GLY dehydration and ACL hydrogenation respectively and can be related to temperature using Eq. 5 and Eq. 6.

$$k_1 = 1.54 \times 10^7 \exp \frac{-86560 \text{kJ/kmol}}{RT} \quad \text{Eq. 5}$$

$$k_2 = 716 \exp \frac{-57800 \text{kJ/kmol}}{RT} \quad \text{Eq. 6}$$

Crude glycerol consists of 70 wt% of glycerol and the balance is water. Crude glycerol with a flow rate of 1 kmol/hr is mixed with 5 kmol/hr of hydrogen before entering a plug flow reactor with the diameter of 5 cm and length of 200 cm. The feed temperature and pressure are 200 °C and 1 bar respectively. The reactor is operated at the same condition with the feed. Simulate the reaction using NRTL-RK method and the reaction kinetics given.

- a) Determine the conversion of GLY.
- b) It is desired to achieve 90% reaction conversion in the reactor. Determine the reactor length required to achieve this reaction conversion.
- c) Determine the number of tubes (same dimension with the reactor in (a)) required in a multi-tube reactor for the production of 50,000 tons/operational year of PDIOL. Verify the answer through the simulation. (one operational year is= 330 days x 24 hrs= 7920 hrs)

Problem Analysis

- Components involved: GLY, water, ACL, H₂, PDIOL
- Property method: NRTL-RK
- Feed information:
 - Number of inlet feed: 2
 - Feed 1: 1 kmol/hr Crude glycerol (70 wt% glycerol, 30 wt% water)
 - Feed 2: 5 kmol/hr Hydrogen (100% hydrogen)

Reactor information:

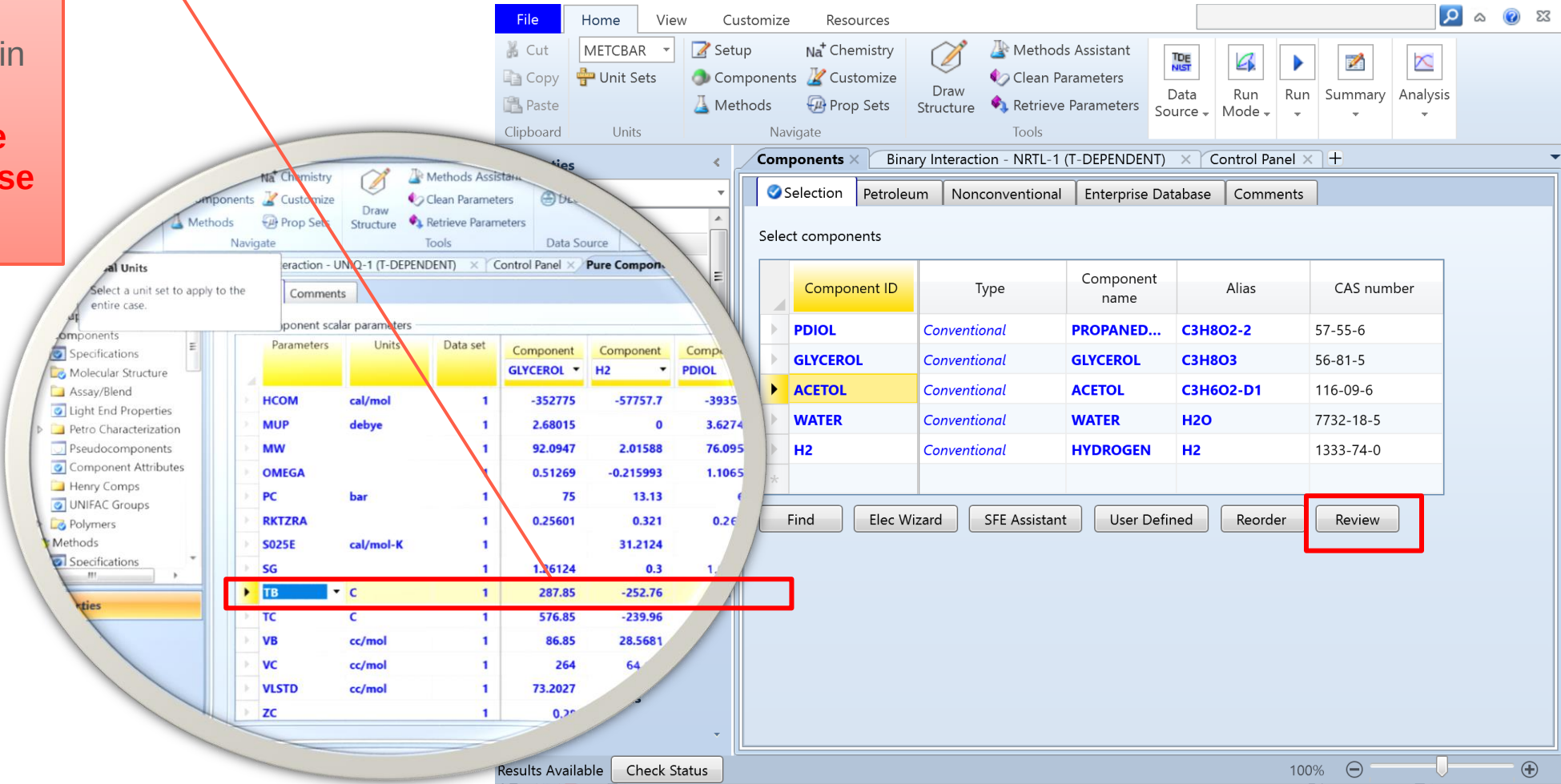
- Type of reactor model:
 - ✓ **Kinetic based model R-Plug as the kinetic info must be included**
 - × R-Yield
 - × R-Stoic as the conversion/extent of reaction is not given
 - × Equilibrium based model such as R-Equil and R-Gibbs
- Valid phase in the reactor: vapor liquid because the boiling point of glycerol is more than 200°C at 1 bar
- Reactor operating condition: same as feed

Problem Solving

a) Determine the conversion of GLY

$$\text{Conversion, } X = \frac{F_{A0} - F_A}{F_{A0}}$$

Quick check on
 the **boiling
 point, TB at 1
 atm** of the
 components in
 the list to
**confirm the
 reaction phase**



The screenshot shows the Aspen Plus software interface. A red circle highlights the 'Component scalar parameters' table, which lists various properties for the selected components. A red rectangle highlights the 'TB' (Boiling Point) row, showing values for Glycerol and Acetol. A red arrow points from the text box on the left to the 'TB' row in the table.

Parameters	Units	Data set	Component Glycerol	Component H2	Component PDIOL
HCOR	cal/mol	1	-352775	-57757.7	-3935
MUP	debye	1	2.68015	0	3.6274
MW		1	92.0947	2.01588	76.095
OMEGA		1	0.51269	-0.215993	1.1065
PC	bar	1	75	13.13	
RKTZRA		1	0.25601	0.321	0.26
S02SE	cal/mol-K	1		31.2124	
SG		1	1.26124	0.3	1
TB	C	1	287.85	-252.76	
TC	C	1	576.85	-239.96	
VB	cc/mol	1	86.85	28.5681	
VC	cc/mol	1	264	64	
VLSTD	cc/mol	1	73.2027		
ZC		1	0.2		

The 'Components' panel on the right shows a list of selected components:

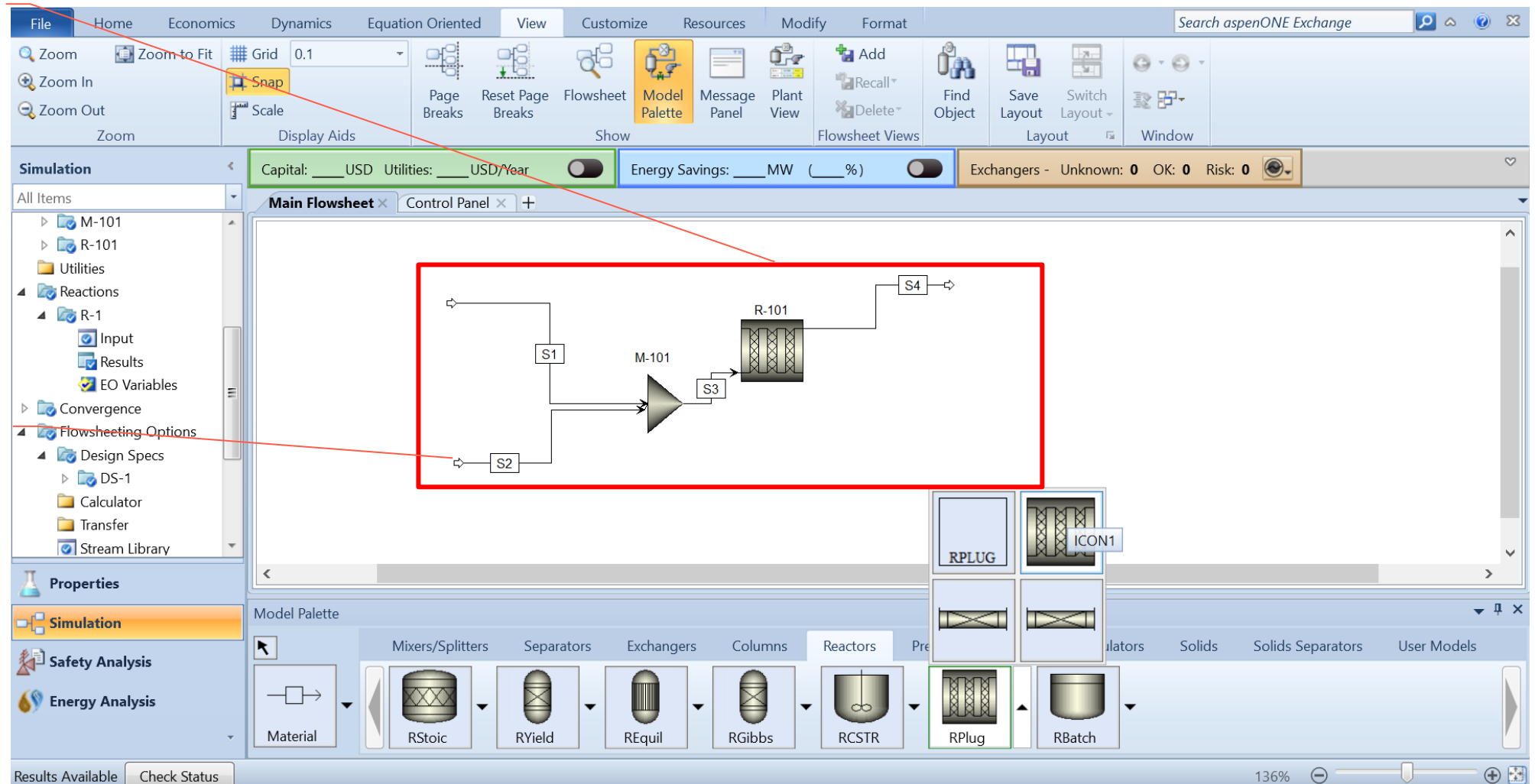
Component ID	Type	Component name	Alias	CAS number
PDIOL	Conventional	PROPANED...	C3H8O2-2	57-55-6
GLYCEROL	Conventional	GLYCEROL	C3H8O3	56-81-5
ACETOL	Conventional	ACETOL	C3H6O2-D1	116-09-6
WATER	Conventional	WATER	H2O	7732-18-5
H2	Conventional	HYDROGEN	H2	1333-74-0

The 'Review' button is highlighted with a red rectangle.

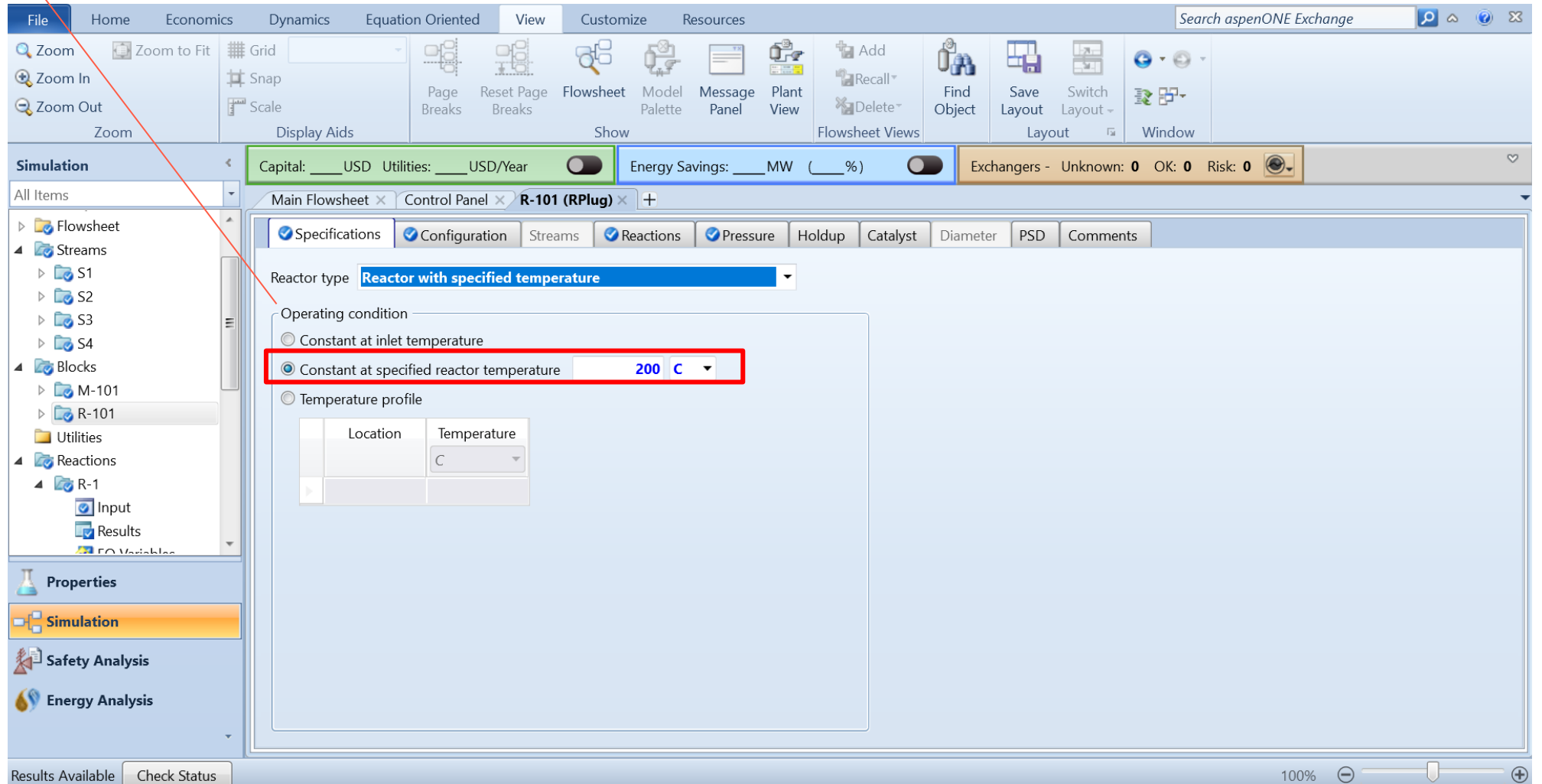
Construct the flowsheet by including the block R-Plug and connecting it with streams

Mixer, M-101 is required as two inlet feeds are now allowed for R-Plug

2 feed streams:
S1- Crude glycerol
S2- Hydrogen



Select 2nd option
 and
 Include the
 reactor
 temperature



File Home Economics Dynamics Equation Oriented View Customize Resources Search aspenONE Exchange

Zoom Zoom to Fit Grid Snap Page Breaks Reset Page Breaks Flowsheet Model Palette Message Panel Plant View Add Recall Delete Find Object Save Layout Switch Layout Window

Simulation Capital: ___ USD Utilities: ___ USD/Year Energy Savings: ___ MW (___%) Exchangers - Unknown: 0 OK: 0 Risk: 0

Main Flowsheet x Control Panel x R-101 (RPlug) x

Specifications Configuration Streams Reactions Pressure Holdup Catalyst Diameter PSD Comments

Reactor type **Reactor with specified temperature**

Operating condition

☐ Constant at inlet temperature

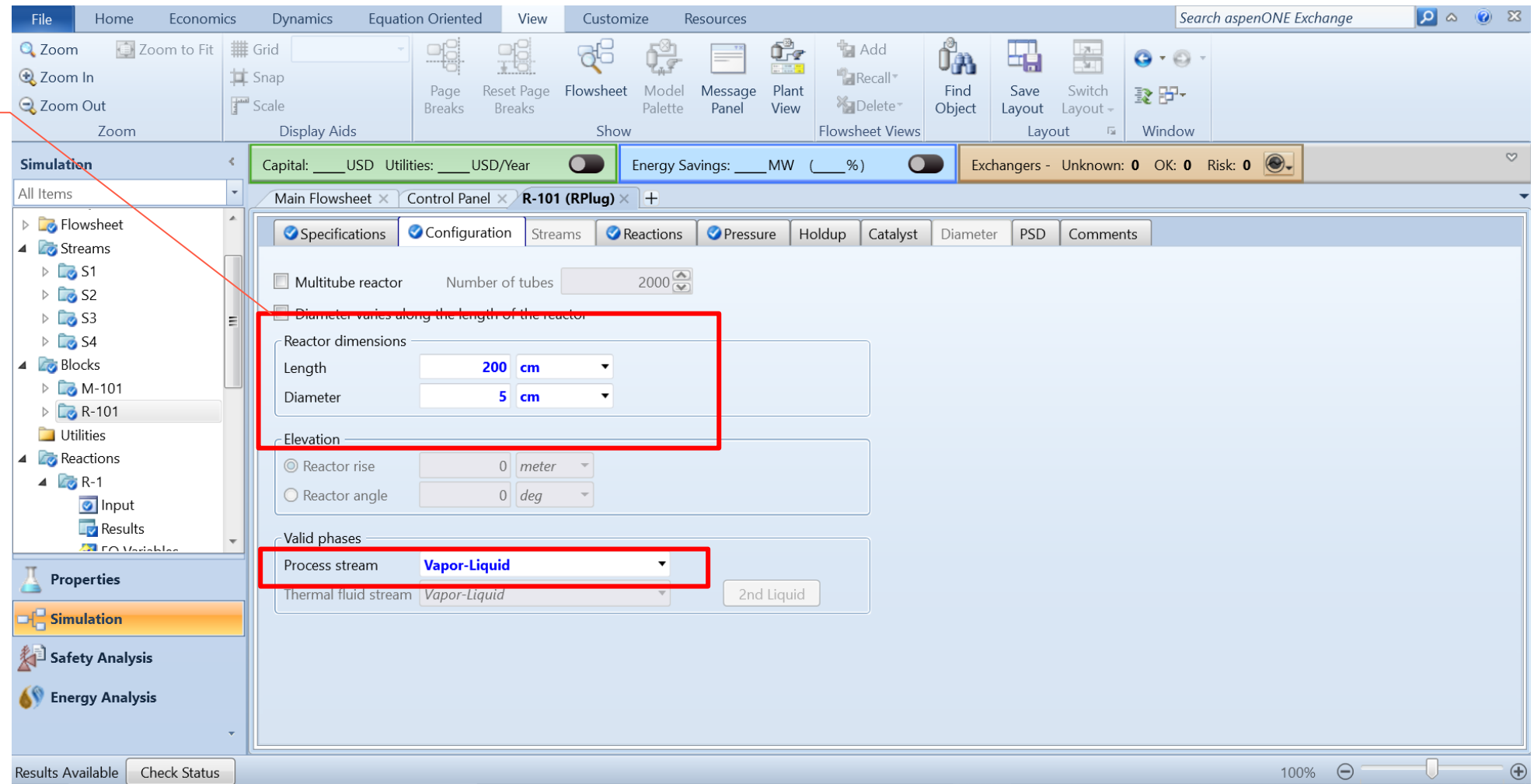
☒ **Constant at specified reactor temperature 200 C**

☐ Temperature profile

Location	Temperature
	C

Results Available Check Status 100%

Include the reactor dimensions



The screenshot displays the Aspen Plus software interface for configuring a reactor (R-101). The left sidebar shows the project hierarchy with 'R-101' selected under 'Blocks'. The main window shows the 'Specifications' and 'Configuration' tabs. The 'Reactor dimensions' section is highlighted with a red box, showing the following settings:

- Length: 200 cm
- Diameter: 5 cm

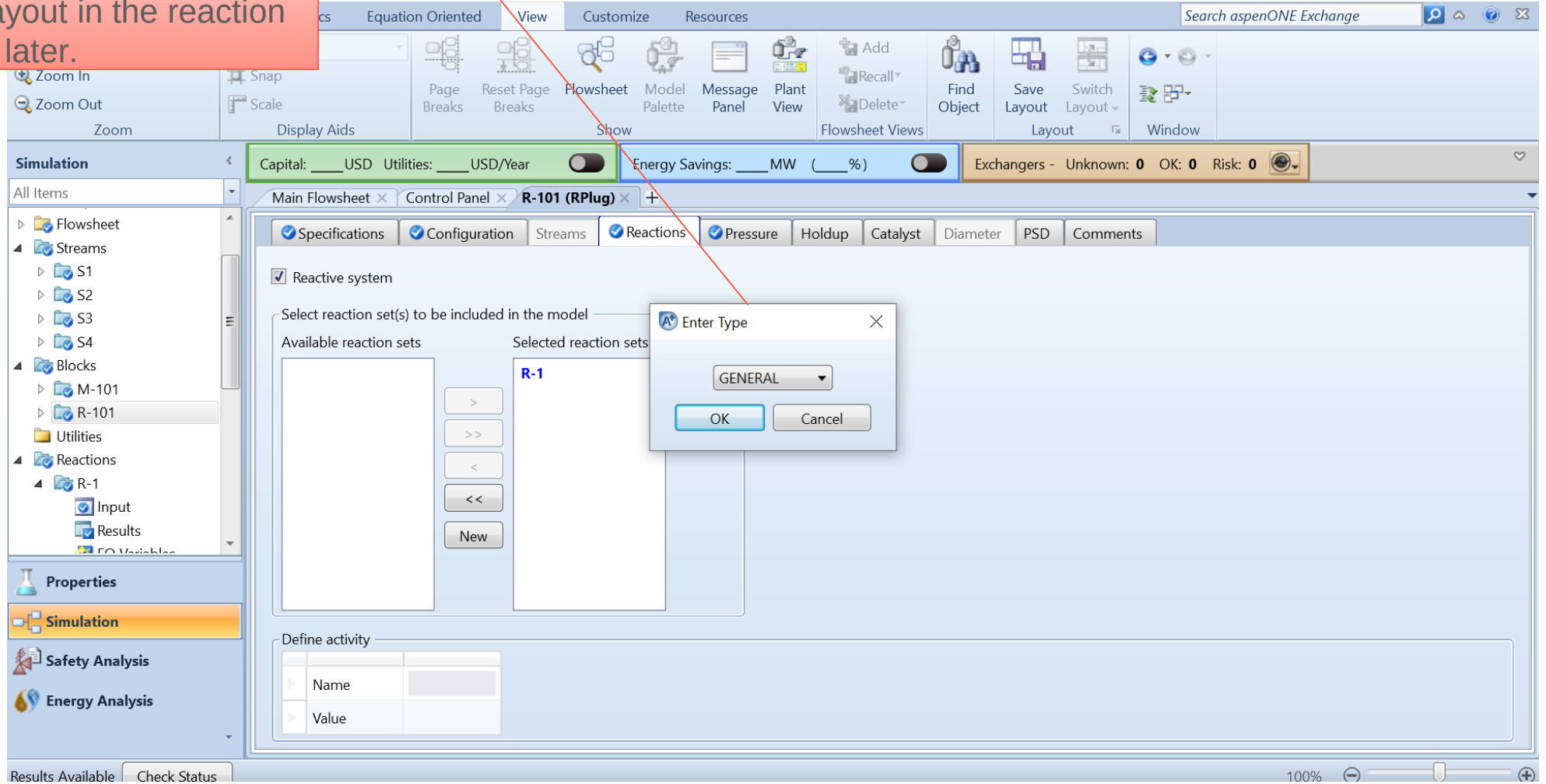
The 'Valid phases' section is also highlighted with a red box, showing the following settings:

- Process stream: Vapor-Liquid
- Thermal fluid stream: Vapor-Liquid

Other visible settings include 'Number of tubes: 2000' and 'Reactor rise: 0 meter'.

Create a chemical reactive system, R1

Make sure GENERAL type is chosen to obtain the desired layout in the reaction setup later.



The screenshot displays the Aspen Plus software interface. The 'Enter Type' dialog box is open, showing 'GENERAL' as the selected reaction type. The background window shows the 'Reactions' tab with 'Reactive system' checked. The 'Selected reaction sets' list contains 'R-1'. The 'Define activity' section at the bottom is empty.

Include the stoichiometry of the chemical reactions for both

1. Glycerol dehydration
2. Acetol hydrogenation

Capital: ___USD Utilities: ___USD/Year Energy Savings: ___MW (___%) Exchangers - Unknown: 0 OK: 0 Risk: 0

B2 (RPlug) x HYDROGEN (MATERIAL) x Main Flowsheet x R-1 (GENERAL) x Reactions x Setup x +

Configuration Kinetic Equilibrium Custom Activity GLHHW Adsorption Kinetics Summary Comments

New Copy Paste Edit Input

No.	Name	Reaction class	Active	Reversible	Stoichiometry	Delete
1	INTER	POWERLAW	☒	☐	GLYCEROL --> ACETOL(MIXED) + WATER(MIXED)	☒
2	MAIN	POWERLAW	☒	☐	ACETOL + H2 --> PDOL(MIXED)	☒

Model Palette: Mixers/Splitters Separators Exchangers Columns Reactors Pressure Changers Manipulators Solids Solids Separators Batch Models User Models

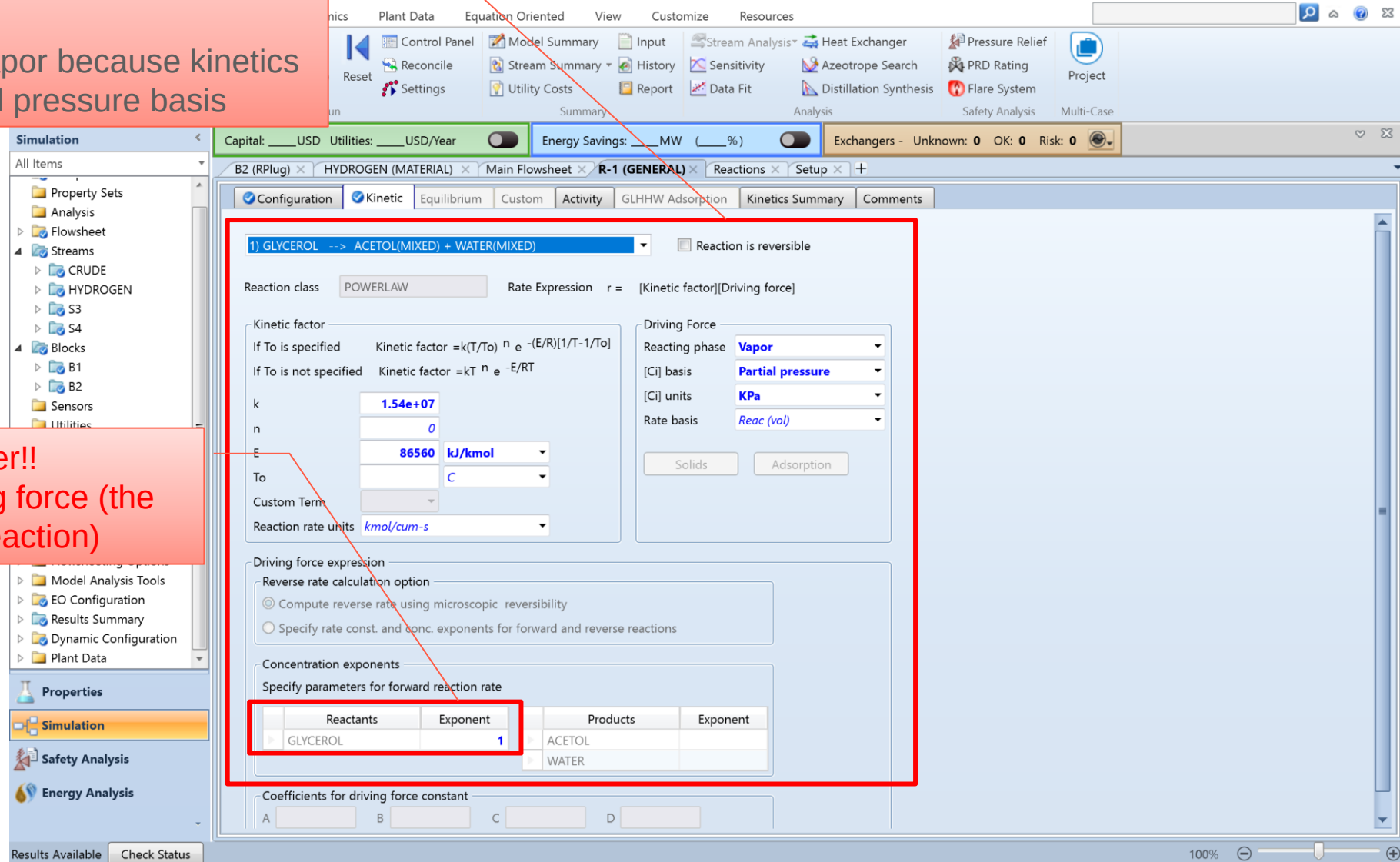
MATERIAL Mixer FSplit SSplit

Results Available Check Status 100%

Include the kinetic parameters for 1st reaction, glycerol dehydration

Reacting phase: vapor because kinetics given in partial pressure basis

Remember!!
Include the driving force (the order of the reaction)



Simulation

Capital: ___USD Utilities: ___USD/Year Energy Savings: ___MW (___%) Exchangers - Unknown: 0 OK: 0 Risk: 0

B2 (RPlug) x HYDROGEN (MATERIAL) x Main Flowsheet x R-1 (GENERAL) x Reactions x Setup x +

Configuration Kinetic Equilibrium Custom Activity GLHHW Adsorption Kinetics Summary Comments

1) GLYCEROL --> ACETOL(MIXED) + WATER(MIXED) ☐ Reaction is reversible

Reaction class POWERLAW Rate Expression $r = [\text{Kinetic factor}][\text{Driving force}]$

Kinetic factor

If T_0 is specified Kinetic factor $= k(T/T_0)^n e^{-(E/R)[1/T - 1/T_0]}$

If T_0 is not specified Kinetic factor $= kT^n e^{-E/RT}$

k 1.54e+07

n 0

E 86560 kJ/kmol

To C

Custom Term

Reaction rate units kmol/cum-s

Driving Force

Reacting phase Vapor

[C] basis Partial pressure

[C] units KPa

Rate basis Reac (vol)

Solids Adsorption

Driving force expression

Reverse rate calculation option

☒ Compute reverse rate using microscopic reversibility

☐ Specify rate const. and conc. exponents for forward and reverse reactions

Concentration exponents

Specify parameters for forward reaction rate

Reactants	Exponent	Products	Exponent
GLYCEROL	1	ACETOL	
		WATER	

Coefficients for driving force constant

A B C D

Results Available Check Status 100%

Include the kinetic parameters for 2nd reaction, acetol hydrogenation

2) ACETOL + H2 --> PDIOL(MIXED)

1) GLYCEROL --> ACETOL(MIXED) + WATER(MIXED)

2) ACETOL + H2 --> PDIOL(MIXED)

Kinetic factor

If To is specified Kinetic factor = $k(T/T_o)^n e^{-(E/R)[1/T-1/T_o]}$

If To is not specified Kinetic factor = $kT^n e^{-E/RT}$

k 716

n 0

E 57800 kJ/kmol

To C

Custom Term

Reaction rate units kmol/cum-s

Driving Force

Reacting phase Vapor

[Ci] basis Partial pressure

[Ci] units KPa

Rate basis Reac (vol)

Driving force expression

Reverse rate calculation option

Compute reverse rate using microscopic reversibility

Specify rate const. and conc. exponents for forward and reverse reactions

Concentration exponents

Specify parameters for forward reaction rate

Reactants	Exponent	Products	Exponent
ACETOL	1	PDIOL	
H2	1		

Coefficients for driving force constant

A B C D

Check the stream results
 Calculate the conversion using:

$$X_{GLY} = \frac{\text{Molar flowrate of GLY in the inlet stream} - \text{Molar flowrate of GLY in the outlet stream}}{\text{Molar flowrate of GLY in the inlet stream}}$$

$$= \frac{0.313 - 0.115}{0.313} = 63.3\% \text{ (Answer for Question (a))}$$

Simulation

Capital: ___USD Utilities: ___USD/Year Energy Savings: ___MW (___%) Exchangers - Unknown: 0 OK: 0 Risk: 0

Main Flowsheet Control Panel R-101 (RPlug) - Stream Results (Default)

Material	Heat	Load	Vol.% Curves	Wt.% Curves	Petroleum	Polymers	Solids	Sum: 0.428277
Units S3 S4								
Average MW					8.55179	8.51672		
- Mole Flows kmol/hr								
GLYCEROL					0.313394	0.114883		
H2					5	4.82619		
PG					0	0.173808		
EG					0	0		
H2O					0.686606	0.885117		
ACETOL					0	0.0247026		
+ Mole Fractions								
+ Mass Flows kg/hr								
					51.3107	51.3107		
+ Mass Fractions								
Volume Flow					3573.84	3950.45		
+ Vapor Phase								
+ Liquid Phase								

Results Available Check Status 100%

Problem Solving

- b) It is desired to achieve 90% reaction conversion in the reactor. Determine the reactor length required to achieve this reaction conversion.
- Vary the reactor length manually until the conversion of glycerol achieve 0.9, or
 - Adopting Design Spec function, set the range of Varying Parameter and the Targeted Conversion

Achieving Process Objectives using Design Specifications

Design Specification

- Allow user to set the value of a calculated flowsheet quantity to a particular value
- Objective is achieved by manipulating a specified input variable
- Located in the **Navigation Pane | Flowsheeting Options | Design Specs** or the **Manipulators** tab of the **Model Palette**
- Design specifications change the simulation results
- **Design Spec** block can be placed on the PFD using icons on the **Manipulators** tab of the **Model Palette**
 - Dashed connection lines will indicate the unit operation models and streams affected by these blocks
 - Display can be toggled on/off from the **File | Options | Flowsheet** dialog box

Steps for Using Design Specifications

1. Define measured (sampled) variables
2. Specify objective function (**Spec**) and Goal (**Target**)
3. Set tolerance for objective function
4. Specify manipulated (varied) variable
5. Specify range of manipulated (varied variable)

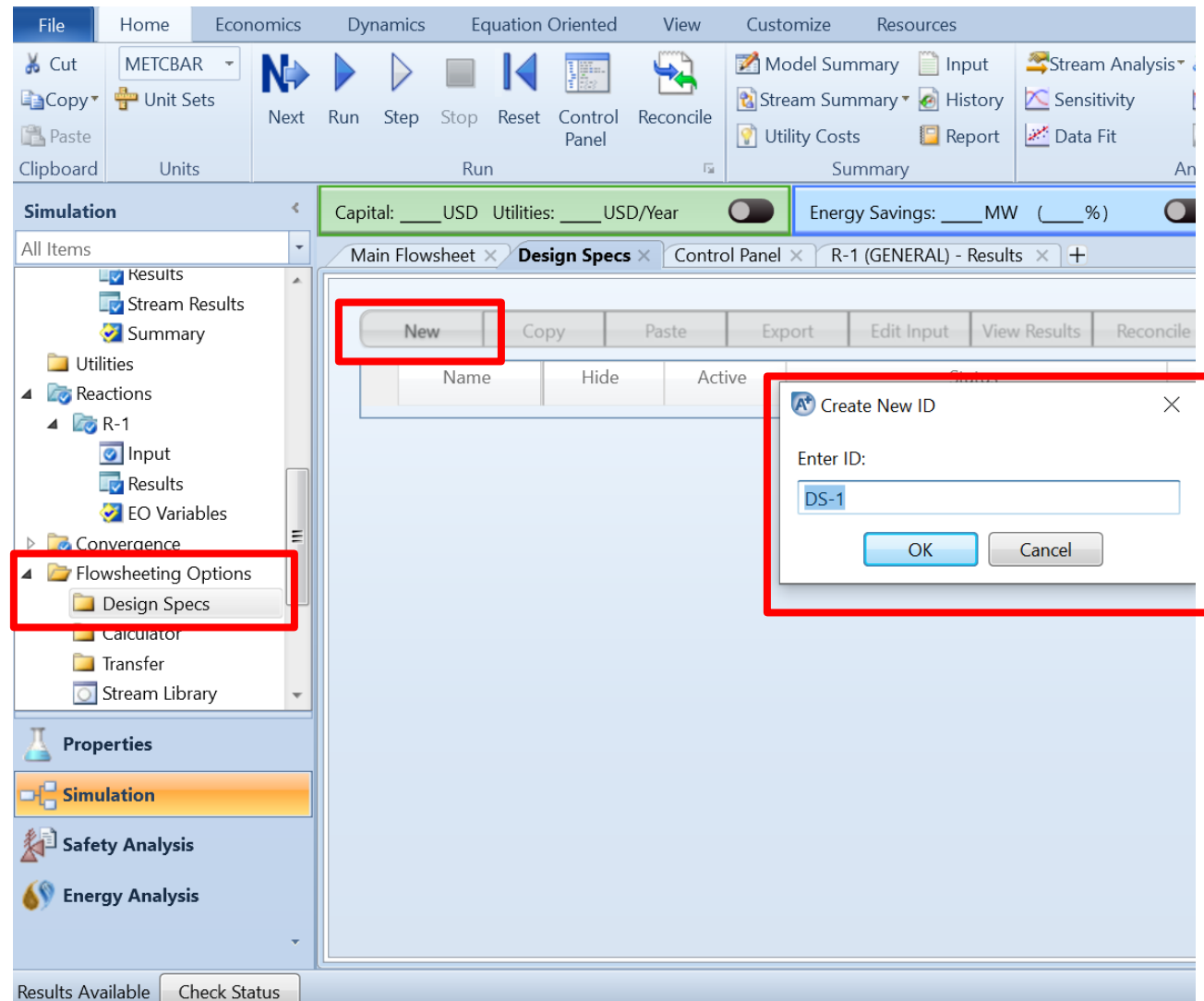
Design Specification

- Only quantities that have been input to the flowsheet should be manipulated
- The calculation performed by a design spec are iterative; providing a good estimate for the manipulated variable will help the design spec converge in fewer iterations; specially for larger flowsheet with several design spec
- Design spec are an iterative calculation that requires a convergence block
- Flowsheet with recycles will require iterative calculations to solve and associated convergence block
- If both Design specs and recycles are involved, typically the convergence block for the Design specs is “Nested” or converged inside of the convergence block for the recycle stream (Default)
- For some flowsheet the Design Spec convergence block needs to be converged outside of the recycle stream convergence block
- Secant Method is the default method for converging single design specification

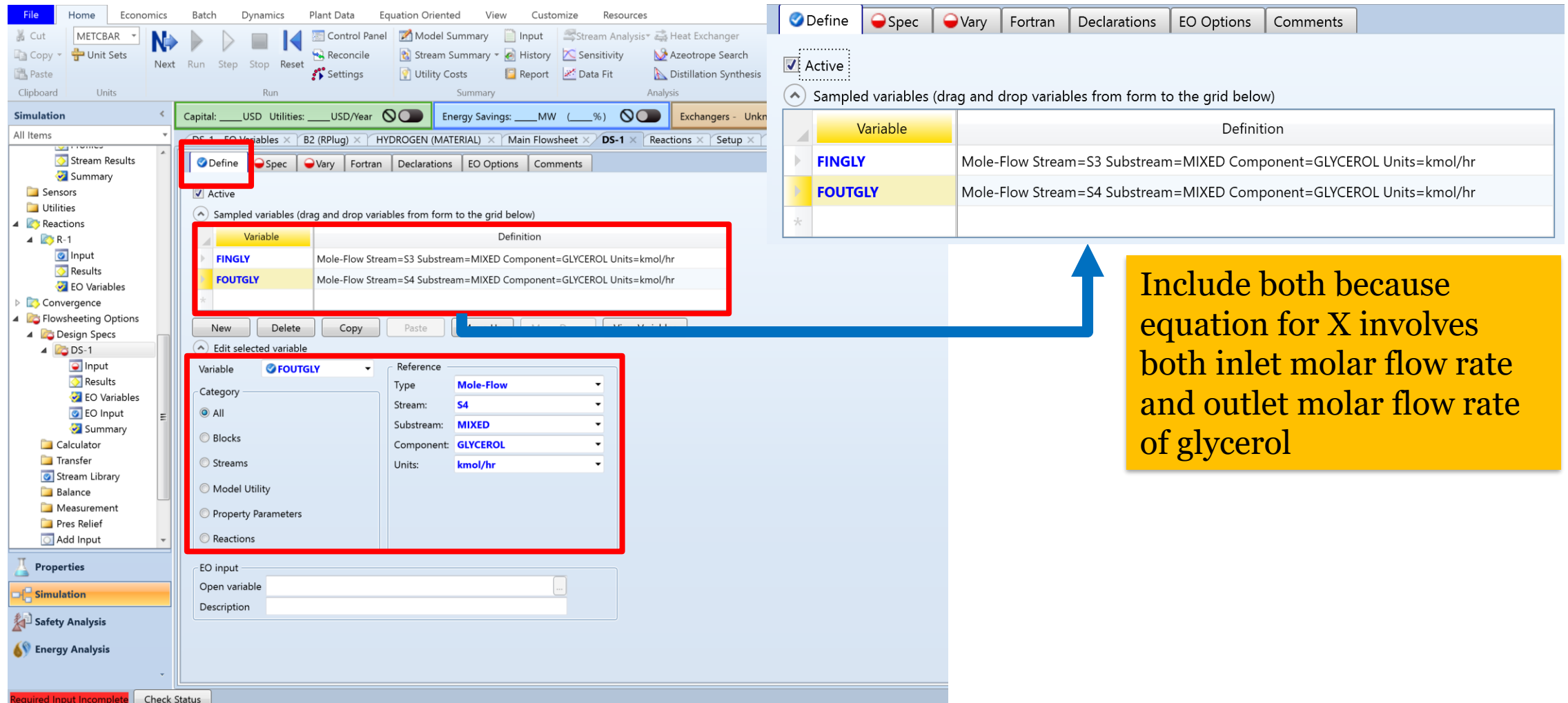
Design Specification

- If a design-spec does not converge:
 - Check the lower and upper bound of the manipulated variable
 - Verify that a solution exists within the bounds specified for the manipulated variable, perhaps by performing a sensitivity analysis
 - Ensure the MV does affect the value of the sample variables
 - Provide better estimate for the value of MV
 - Loosening the tolerance to help convergence
 - Make sure that the objective function does not have a flat region within the range of the manipulated variable
 - Change the characteristic of the convergence block associated with the design-spec

- Select design specs under Flowsheeting Options> create a new design specs analysis



- Select design specs under Flowsheeting Options> create a new design specs analysis> **define the variable for the targeted parameters**



The screenshot displays the Aspen Plus software interface, specifically the 'Define' tab for design specifications. The 'Active' section shows a table with variables FINGLY and FOUTGLY. The 'Edit selected variable' section shows the configuration for FOUTGLY.

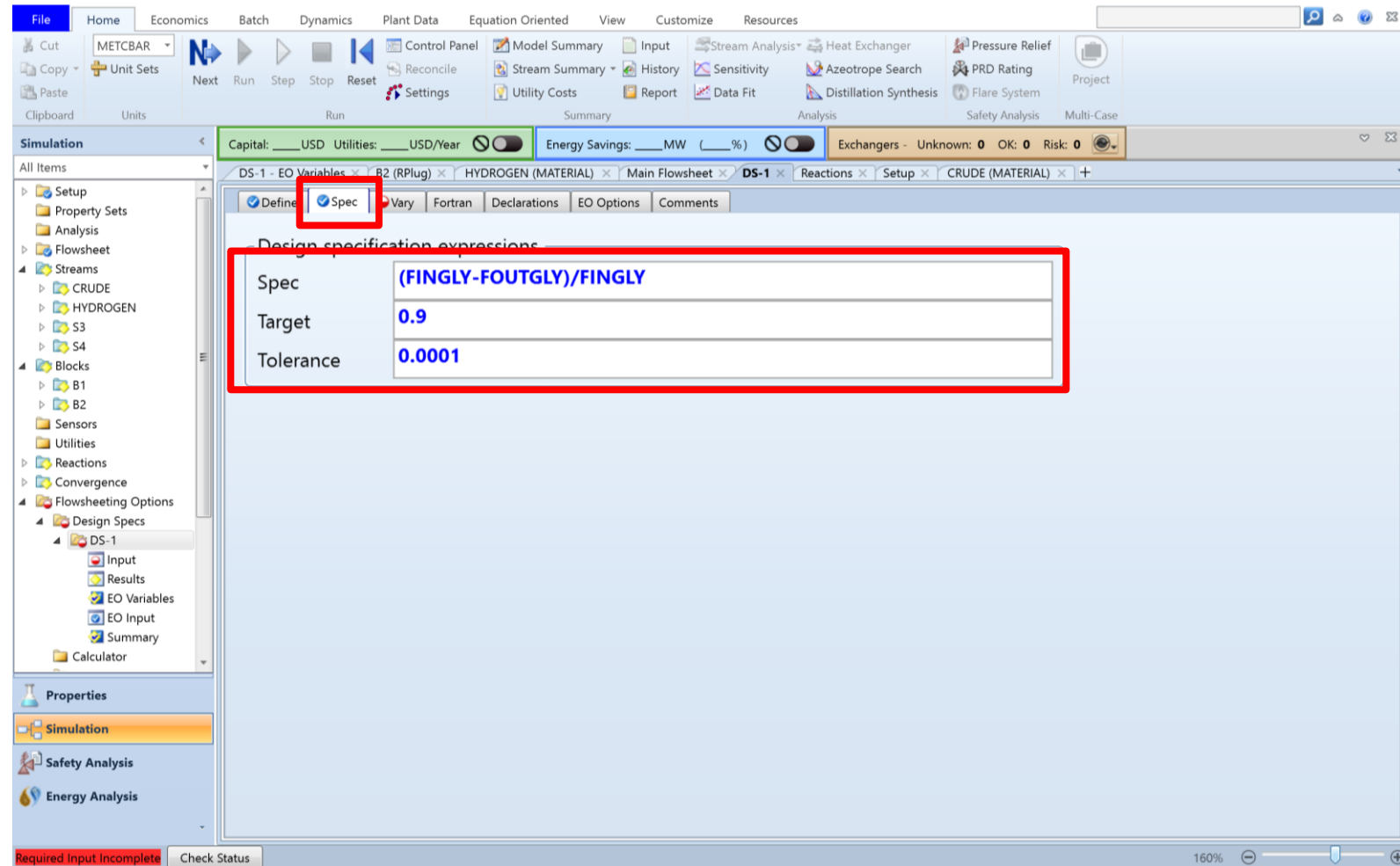
Variable	Definition
FINGLY	Mole-Flow Stream=S3 Substream=MIXED Component=GLYCEROL Units=kmol/hr
FOUTGLY	Mole-Flow Stream=S4 Substream=MIXED Component=GLYCEROL Units=kmol/hr

Variable	Reference
FOUTGLY	Type: Mole-Flow Stream: S4 Substream: MIXED Component: GLYCEROL Units: kmol/hr

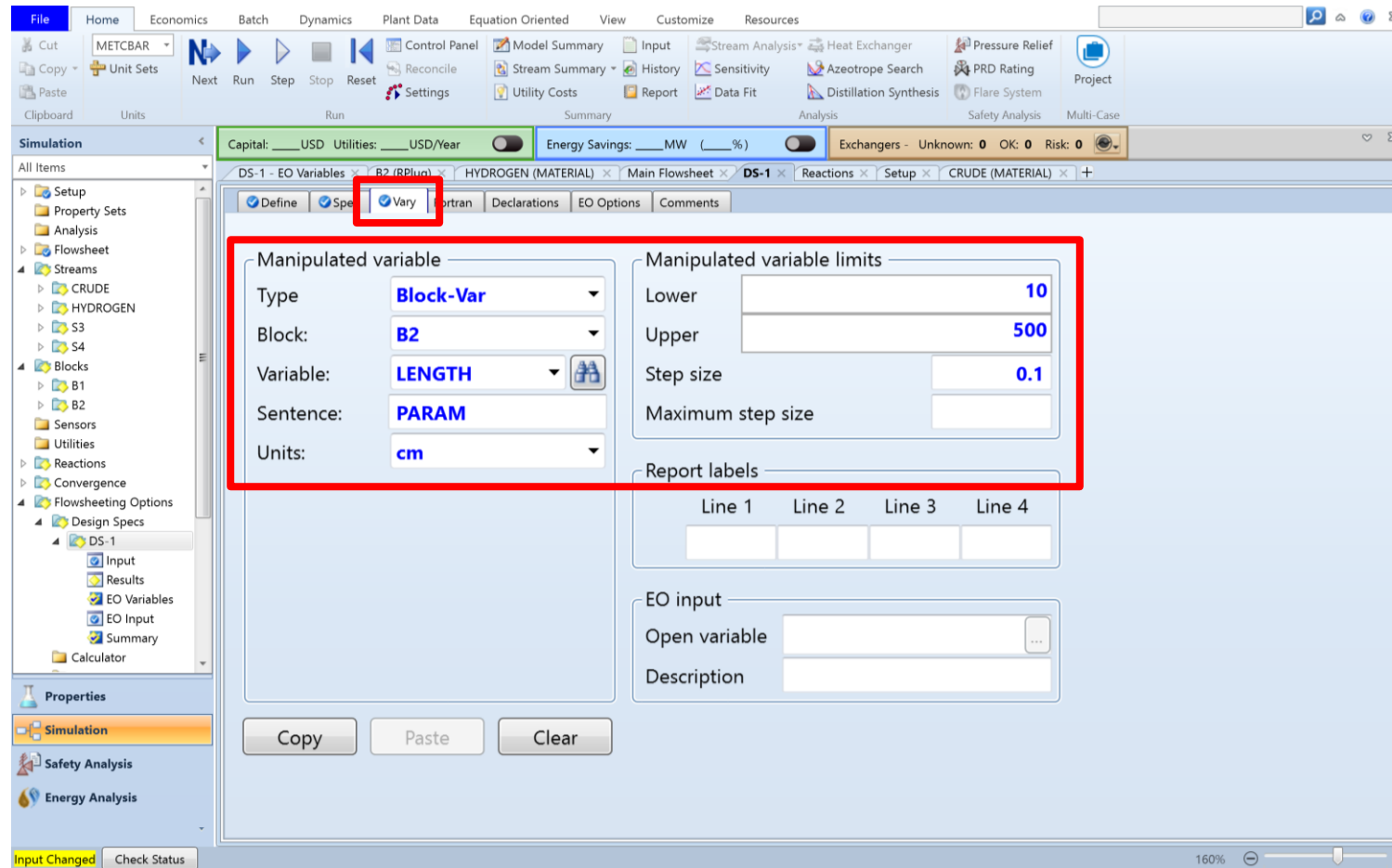
EO input
Open variable
Description

Include both because equation for X involves both inlet molar flow rate and outlet molar flow rate of glycerol

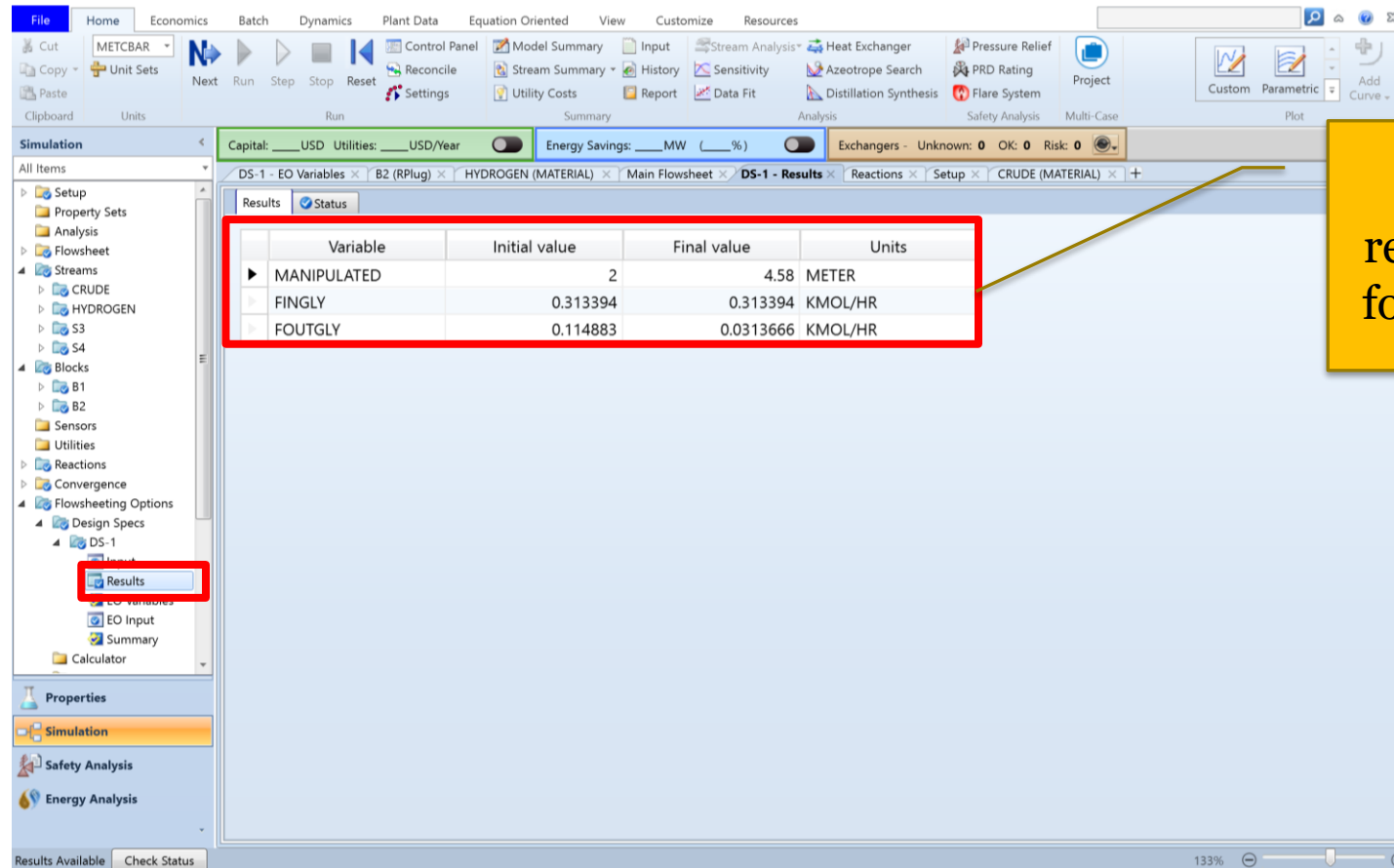
- Select design specs under Flowsheeting Options> create a new design specs analysis> define the variable for the targeted parameters> **set the spec of the targeted**



- Select design specs under Flowsheeting Options> create a new design specs analysis> define the variable for the targeted parameters> set the spec of the targeted conversion> **include varying parameter**



- Select design specs under Flowsheeting Options> create a new design specs analysis> define the variable for the targeted parameters> set the spec of the targeted conversion> include varying parameter> **run simulation and check results**



Variable	Initial value	Final value	Units
MANIPULATED	2	4.58	METER
FINGLY	0.313394	0.313394	KMOL/HR
FOUTGLY	0.114883	0.0313666	KMOL/HR

The reactor length required= 4.58 meter for achieving X of 0.9

Problem Solving

- c) Determine the number of tubes (same dimension with the reactor in (a)) required in a multi-tube reactor for the production of 50,000 tons/operational year of PDIOL. Verify the answer through the simulation. (one operational year is= 330 days x 24 hrs= 7920 hrs)

Change the no. of hours
in an operational year.

The screenshot displays the Aspen Plus software interface. On the left, the 'Simulation' tree shows the 'Setup' option under the 'B2' block, highlighted with a red box. A red arrow points from this box to the 'Global settings' panel on the right. In the 'Global settings' panel, the 'Operational year' is set to '7920 hr', which is also highlighted with a red box. Other settings visible include 'Global unit set' as 'METCBAR', 'Input mode' as 'STEADY-STATE', 'Stream class' as 'CONVEN', 'Flow basis' as 'Mole', 'Ambient pressure' as '1.01325 bar', 'Ambient temp.' as '10 C', 'Valid phases' as an empty dropdown, and 'Free water' as 'No'.

Production rate in ONE
reactor tube: 115.467
tons/oper-year

No. of tubes required to
produce 50,000 tons/oper-
year= $50000/115.467=$
434 tubes (round up)

Capital: _____ USD Utilities: _____ USD/Year Energy Savings: _____ MW (____%)

Main Flowsheet Control Panel R-101 (RPlug) - Stream Results (Default)

Material	Units	S3	S4
Average MW		8.55179	8.51672
+ Mole Flows	kmol/hr	6	6.0247
+ Mole Fractions			
- Mass Flows	kg/hr	51.3107	51.3107
GLYCEROL	kg/hr	28.8619	10.5801
H2	kg/hr	10.0794	9.72902
PDIOI	tons/oper-year	0	115.467
EG	kg/hr	0	0
H2O	kg/hr	12.3694	15.9456
ACETOL	kg/hr	0	1.82996
+ Mass Fractions			
Volume Flow	l/min	3573.84	3950.45
+ Vapor Phase			
+ Liquid Phase			

Let's verify the answer

The feed flowrate should also be 434 times higher as the feed is for 434 reactor tubes

Capital: ___USD Utilities: ___USD/Year Energy Savings: ___MW (___%) Exchangers -

Main Flowsheet x Control Panel x S1 (MATERIAL) x

☒ Mixed ☐ Solid ☐ NC Solid Flash Options EO Options Costing Comments

Specifications

Flash Type Temperature Pressure

State variables

Temperature 200 C

Pressure 1 bar

Vapor fraction

Total flow basis Mole

Total flow rate 434 kmol/hr

Solvent

Reference Temperature

Volume flow reference temperature C

Component concentration reference temperature C

Composition Mass-Frac

Component	Value
GLYCEROL	0.7
H2	
PDOL	
EG	
H2O	0.3
ACETOL	
Total	1

The feed flowrate
 should also be 434
 times higher as the
 feed is for 434
 reactor tubes

File Home Economics Dynamics Equation Oriented View Customize Resources

Cut Copy Paste Clipboard METCBAR Unit Sets Units

Next Run Step Stop Reset Control Panel Reconcile

Model Summary Stream Summary Utility Costs Input History Report Stream Analysis Sensitivity Data Fit Heat Exchanger Azeotrope Search Distillation Synthesis

Simulation

All Items

Analysis

Flowsheet

Streams

S1

S2

S3

S4

Blocks

M-101

R-101

Setup

Convergence

Report

User Subroutine

Dynamic

Properties

Simulation

Safety Analysis

Energy Analysis

Capital: ___ USD Utilities: ___ USD/Year Energy Savings: ___ MW (___%) Exchangers -

Main Flowsheet Control Panel S2 (MATERIAL)

Mixed CI Solid NC Solid Flash Options EO Options Costing Comments

Specifications

Flash Type Temperature Pressure

State variables

Temperature 200 C

Pressure 1 bar

Vapor fraction

Total flow basis Mole

Total flow rate 2170 kmol/hr

Solvent

Reference Temperature

Volume flow reference temperature C

Component concentration reference temperature C

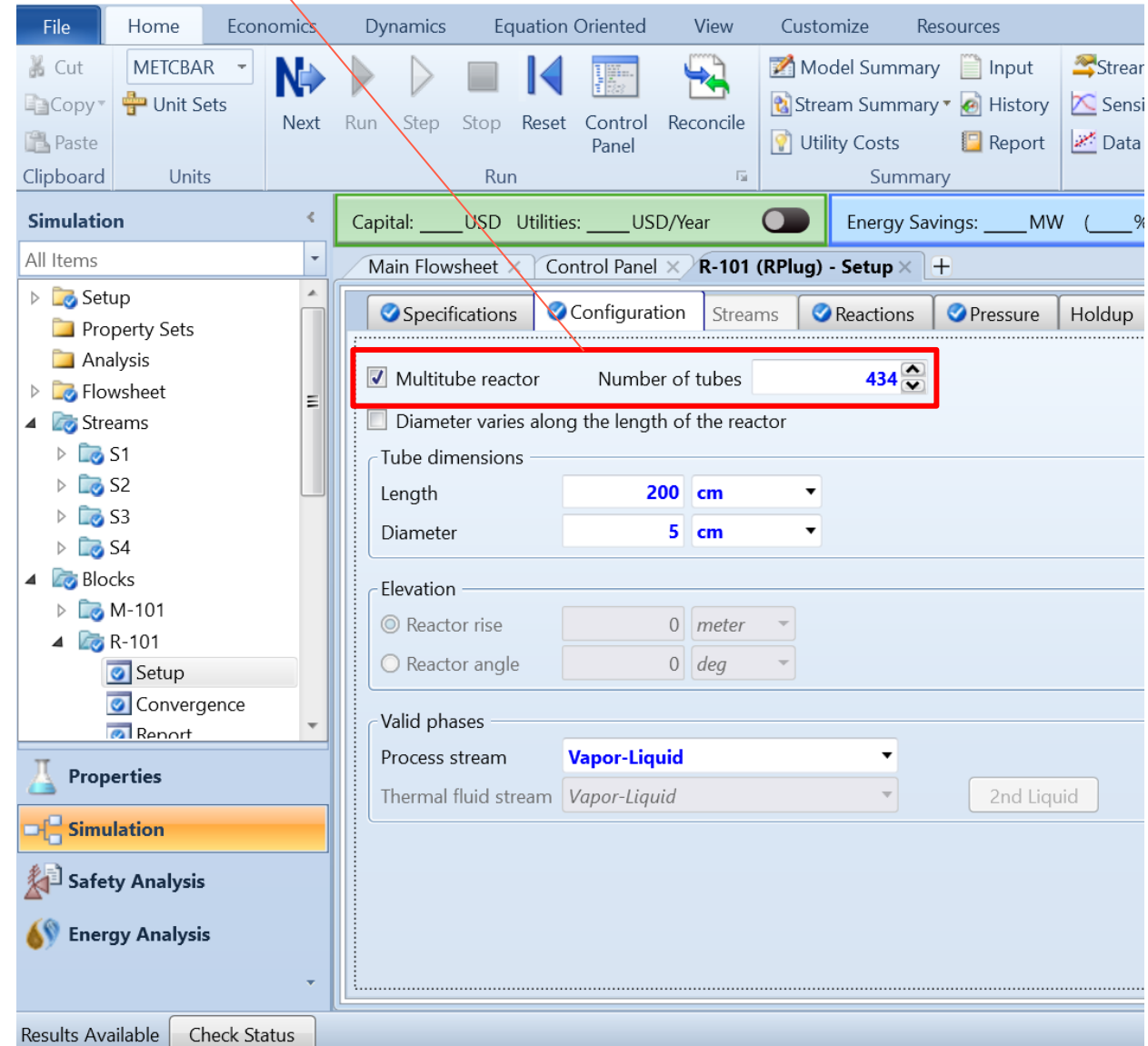
Composition

Mass-Frac

Component	Value
GLYCEROL	
H2	1
PDIOL	
EG	
H2O	
ACETOL	
Total	1

Results Available Check Status

Check the multitube reactor and include 434 tubes



The screenshot displays the Aspen Plus software interface for configuring a reactor. The 'R-101 (RPlug) - Setup' window is open, showing the 'Specifications' tab. A red box highlights the 'Multitube reactor' checkbox, which is checked, and the 'Number of tubes' field, which is set to 434. A red arrow points from the text box on the left to this field. Other visible settings include 'Length' at 200 cm and 'Diameter' at 5 cm. The 'Valid phases' section shows 'Vapor-Liquid' selected for the process stream. The left sidebar shows the project hierarchy with 'R-101' selected under 'Blocks'.

File Home Economics Dynamics Equation Oriented View Customize Resources **Stream Summary**

Default Full  ☒ Total ☒ Mole ☒ Mole
☒ Substreams ☒ Mass ☒ Mass
☒ Phases ☒ Volume ☒ Mass
 General Options Stream Summary Options Flows Composition Property Sets

Save Save as New Show Child Hierarchy Streams Stream Group Template Discard Change

Simulation Capital: ___ USD Utilities: ___ USD/Year Energy Savings: ___ MW (___%)

All Items Main Flowsheet Control Panel **R-101 (RPlug) - Stream Results (Default)**

Material	Heat	Load	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids
		Units		S3	S4		
+ Mole Flows		kmol/hr		2604	2614.72		
+ Mole Fractions							
- Mass Flows		kg/hr		22268.9	22268.9		
GLYCEROL		kg/hr		12526.1	4591.77		
H2		kg/hr		4374.46	4222.4		
PDIOL		tons/oper-year		0	50112.7		
EG		kg/hr		0	0		
H2O		kg/hr		5368.32	6920.4		
ACETOL		kg/hr		0	794.202		
+ Mass Fractions							
Volume Flow		l/min		1.55105e+06	1.71449e+06		
+ Vapor Phase							
+ Liquid Phase							
<add properties>							

Results Available Check Status

Run simulation and check production rate.

50123 tons/oper-year
VERIFIED

Exercise 4.1

Answers

	RStoic	RGibbs	RPlug	RCSTR
Amount of Ethyl Acetate produced (kmol/hr)	130.613	127.217	123.785	119.932
Mass fraction Ethyl Acetate in product stream	0.566	0.552	0.537	0.52
Heat duty (Gcal/hr)	-0.003697064604	-0.0033689555532	-0.0030979576728	-0.0028589477796

4.2 Flowsheet Convergence

Convergence

- Convergence in Aspen Plus is an iterative process consisting of making guesses for tear streams and then comparing the calculated stream values with the guessed values.
- If these values are equivalent within a certain tolerance, then the simulation has successfully converged.
- To evaluate convergence, determine the:
 - Calculation sequence
 - Tear stream
 - Number of iterations to solution
 - Pattern of err/tol value
 - Convergence method used
- Everything you need to know in evaluating the convergence status is written to the **Control Panel**

What is Tear Stream

- A tear stream is a stream inside a recycle loop
- Aspen Plus either begins with the user's initial estimate for this stream or, if no estimate is provided, it starts with an empty, zero-flow stream
- The recycle is solved iteratively, and after working around the loop, Aspen Plus has a new calculated value for the tear stream
 - It compares the new value with the old value and make a new estimate of that stream
 - When the difference between new and old values is within a specified tolerance then the tear stream is converged

Convergence Block

- Each design specification and tear stream has an associated convergence block
- Convergence block determine how guesses for a tear stream or design specification manipulated variable are updated from the iteration to iteration
- Aspen Plus defined convergence block names begin with the character “\$”
- User convergence blocks can be specified in the **Navigation Pane** under **Convergence | Convergence....**
- Default tear convergence setting can be specified under **Convergence | Options | Defaults**

Tolerance

- Tolerance: precision required for convergence

$$\text{Tolerance} = \left| \frac{X_{old} - X_{new}}{X_{old}} \right|$$

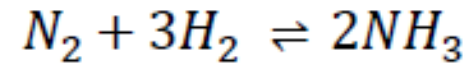
- The default convergence variables are total mole flow, all component mole flow, pressure and enthalpy
- Default block tolerance is 1E-4
- Can be specified locally or globally
- Global flash error tolerance on Setup | Calculation Options | Flash Convergence sheet
- Decrease (tighten) block tolerance to aid in tear stream convergence

Convergence

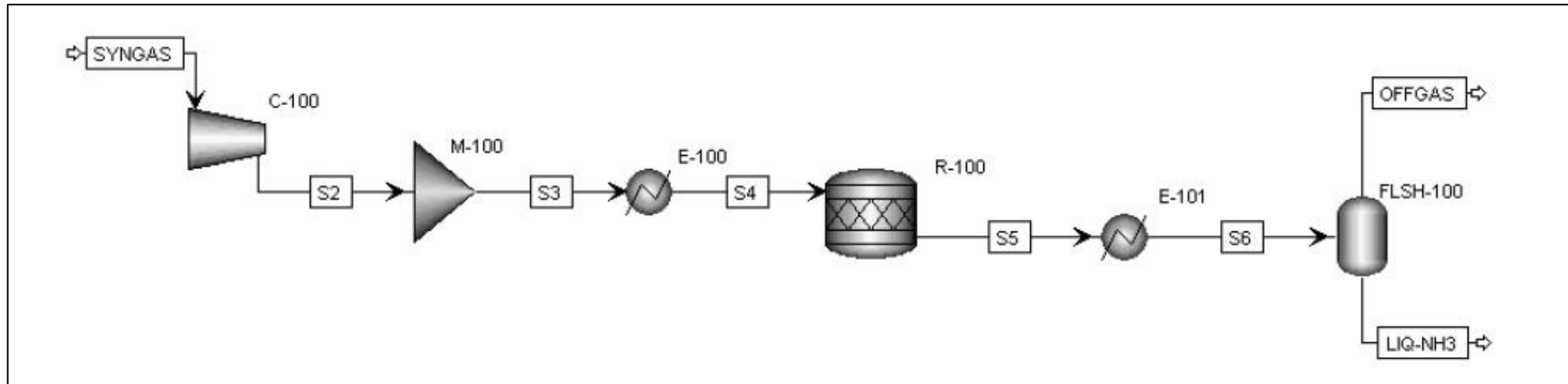
- The following methods are available in Aspen Plus:
 - ✓ Direct substitution
 - ✓ Wegstein
 - ✓ Newton
 - ✓ Secant
 - ✓ Broyden
 - ✓ Sequential quadratic programming (SQP)
- The error due to non-convergence may be attributed to the following reasons:
 - ✓ Inappropriate numerical method for convergence
 - ✓ Insufficient number of iterations

Example 4.3

Ammonia is produced by reacting nitrogen from air with hydrogen. Hydrogen is usually obtained from steam reformation of methane, and nitrogen is obtained from deoxygenated air. The chemical reaction is shown below:



Our goal is to simulate the production of ammonia using Aspen Plus.



SYNGAS feed condition:

- $T = 553.15 \text{ K}$, $P = 26.17 \text{ atm}$
- Component Mole-Flow: $\text{H}_2 = 5160 \text{ kmol/hr}$, $\text{N}_2 = 1732 \text{ kmol/hr}$, $\text{CH}_4 = 72 \text{ kmol/hr}$, $\text{AR} = 19 \text{ kmol/hr}$, $\text{CO} = 17 \text{ kmol/hr}$

Compressor C-100

- Isentropic Compressor
- Discharge pressure = 271.4 atm

Mixer M-100

- No pressure drop across the mixer

Heater E-100

- Temperature = 755 K (Outlet temperature)
- Heater with zero pressure drop- enter 0 in the Pressure spec

Reactor R-100

- Reaction temperature and pressure: 755 K, 270 atm
- Valid phases: vapor liquid
- Reaction Stoichiometric Coefficients: $N_2 = -1$, $H_2 = -3$, $NH_3 = +2$
- Specify Product Regeneration: Fractional conversion = 40% of the reactant N_2

Cooler E-101

- Cooling to 300.15 K
- No pressure drop- enter 0 for the Pressure spec

Flash drum FLSH-100

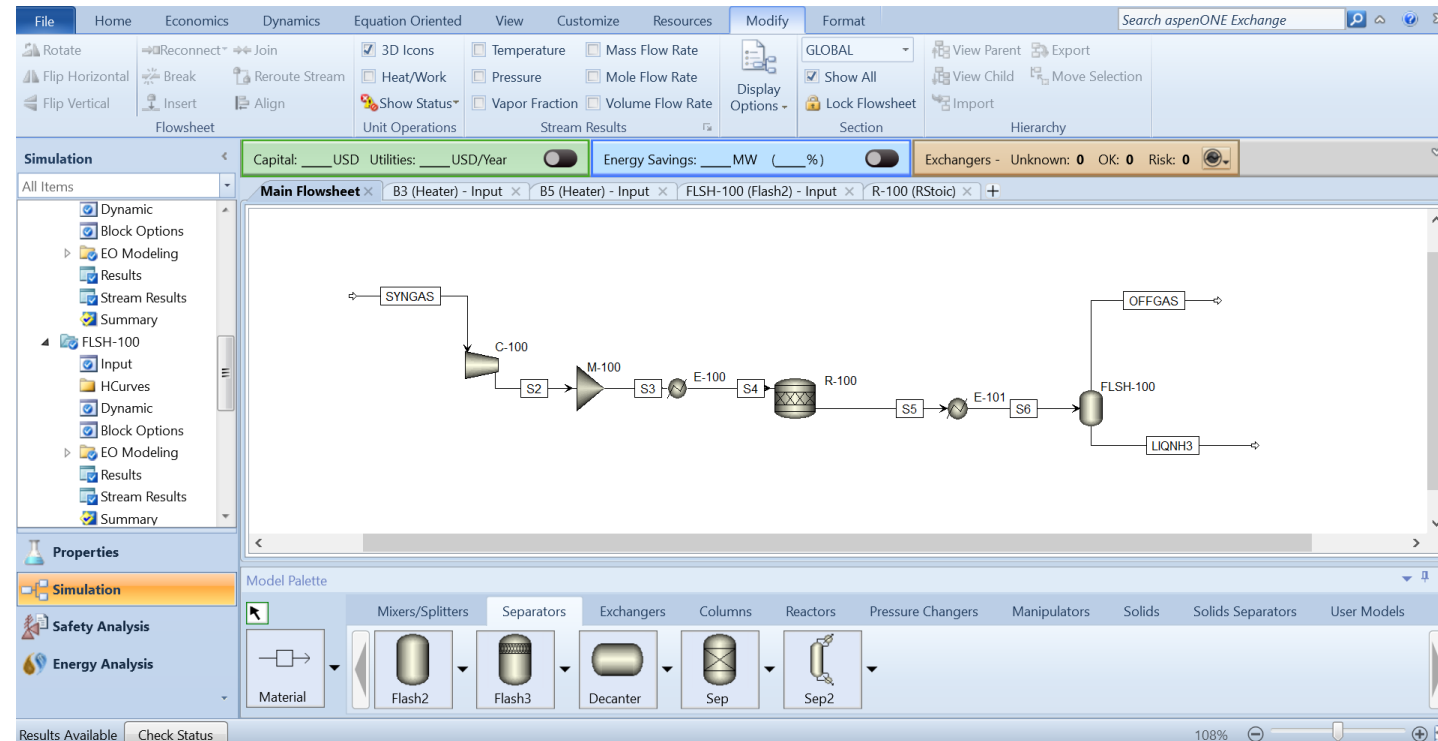
- No additional heat; Heat Duty = 0
- No pressure drop- enter 0 for the Pressure spec

Problem Analysis

- List of components: Feeds and products
H₂, N₂, CH₄, AR, CO and NH₃
- Property method: Redlich-Kwong modification RKS-BM method is chosen. Equation-of-state models provide an accurate description of the thermodynamic properties of the high-temperature, high-pressure conditions encountered in ammonia plants.
- Unit system: Metric Unit
- Equipment model:
 - Heater E-100: HEATER
 - Reactor R-100: R-STOIC
 - Cooler E-101: HEATER
 - Flash drum FLSH-100: FLASH 2

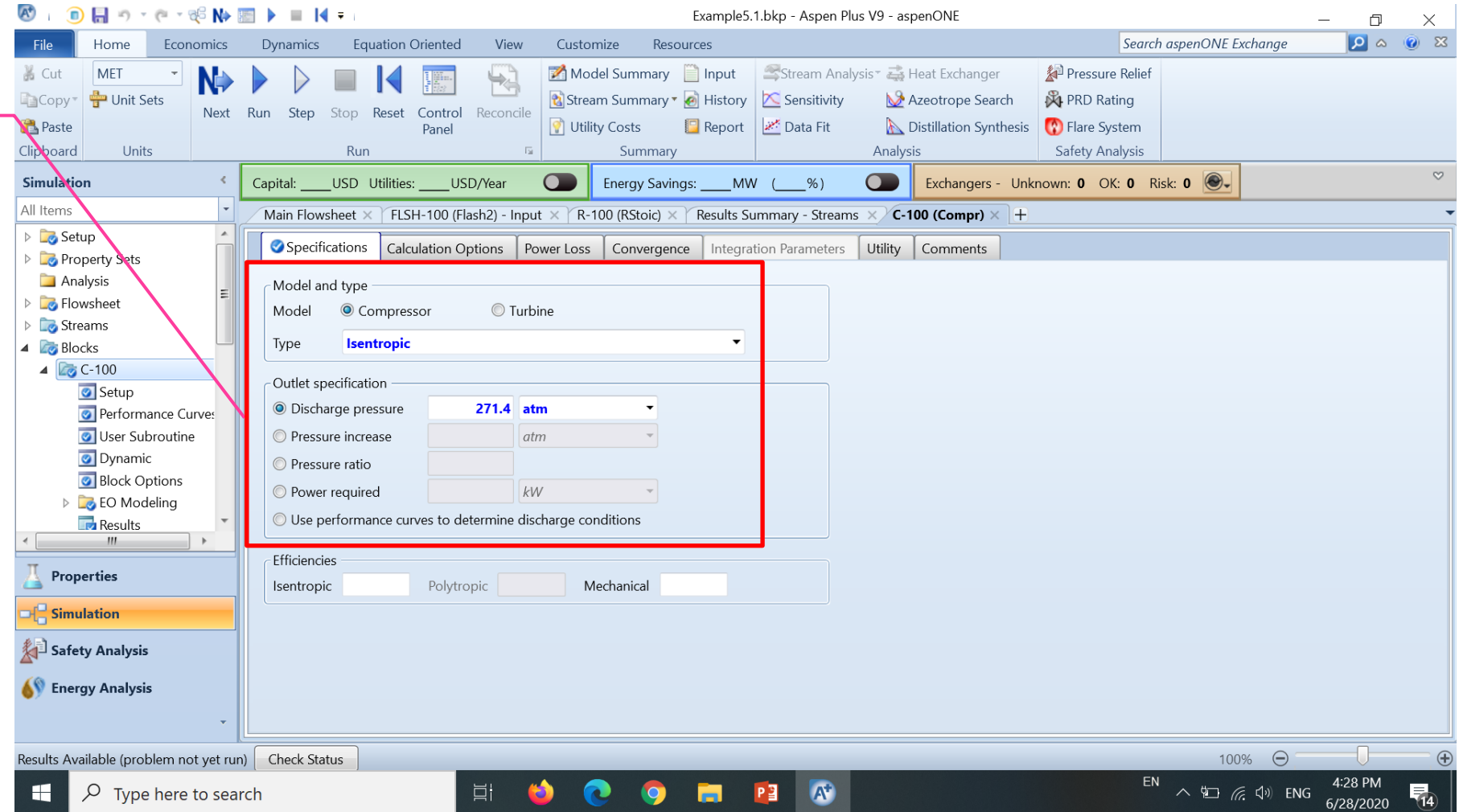
Problem Solving

- Create blank simulation> list the components> select property method> run to generate the properties> shift to simulation tab
- Construct flow sheet> include feed spec> include block spec> run simulation



- Construct flow sheet> include feed spec> include block spec> run simulation

Insert the
 compressor
 specification based
 on the info provided
 in the question



Insert the mixer
 specification based
 on the info provided
 in the question

Example5.1.bkp - Aspen Plus V9 - aspenONE

File Home Economics Dynamics Equation Oriented View Customize Resources

Search aspenONE Exchange

Model Summary Input Stream Analysis Heat Exchanger Pressure Relief
 Stream Summary History Sensitivity Azeotrope Search PRD Rating
 Utility Costs Report Data Fit Distillation Synthesis Flare System
 Summary Analysis Safety Analysis

Simulation Capital: ___ USD Utilities: ___ USD/Year Energy Savings: ___ MW (___%) Exchangers - Unknown: 0 OK: 0 Risk: 0

Main Flowsheet x FLSH-100 (Flash2) - Input x R-100 (RStoic) x Results Summary - Streams x M-100 (Mixer) - Input x

Flash Options Comments

Mixer specifications

Pressure 0 atm

Valid phases Vapor-Liquid

Temperature estimate K

Convergence parameters

Maximum iterations 30

Error tolerance 0.0001

Properties

Simulation

Safety Analysis

Energy Analysis

Results Available (problem not yet run) Check Status

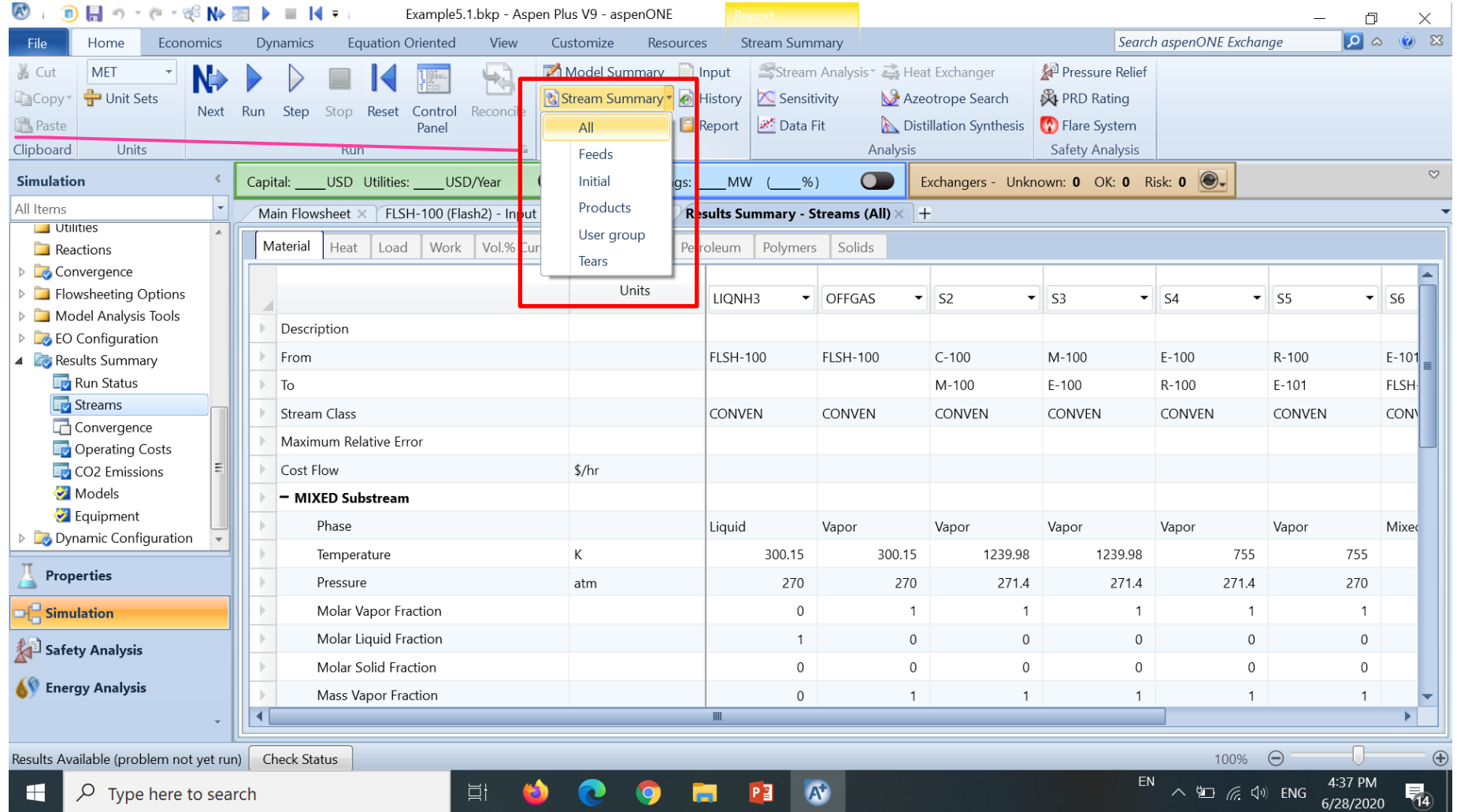
100%

Type here to search

EN 4:34 PM 6/28/2020

- Construct flow sheet> include feed spec> include block spec> **run simulation**

Stream summary
can be generated



Example5.1.bkp - Aspen Plus V9 - aspenONE

File Home Economics Dynamics Equation Oriented View Customize Resources Stream Summary Search aspenONE Exchange

Clipboard Units Run

Simulation Capital: USD Utilities: USD/Year

Main Flowsheet x FLSH-100 (Flash2) - Input

Material Heat Load Work Vol.% Cur

Units

LIQNH3 OFFGAS S2 S3 S4 S5 S6

	LIQNH3	OFFGAS	S2	S3	S4	S5	S6
Description							
From	FLSH-100	FLSH-100	C-100	M-100	E-100	R-100	E-101
To			M-100	E-100	R-100	E-101	FLSH-100
Stream Class	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN
Maximum Relative Error							
Cost Flow							
— MIXED Substream							
Phase	Liquid	Vapor	Vapor	Vapor	Vapor	Vapor	Mixed
Temperature	300.15	300.15	1239.98	1239.98	755	755	
Pressure	270	270	271.4	271.4	271.4	270	
Molar Vapor Fraction	0	1	1	1	1	1	
Molar Liquid Fraction	1	0	0	0	0	0	
Molar Solid Fraction	0	0	0	0	0	0	
Mass Vapor Fraction	0	1	1	1	1	1	

Results Available (problem not yet run) Check Status

100%

EN 4:37 PM 6/28/2020

Stream summary
can be sent to MS
Excel or Flowsheet

Example5.1.bkp - Aspen Plus V9 - aspenONE

File Home Economics Dynamics Equation Oriented View Customize Resources Report

Stream Summary

Search aspenONE Exchange

Send to Excel/ASW
Send to Flowsheet

Capital: ___USD Utilities: ___USD/Year Energy Savings: ___MW (___%) Exchange

Send to Flowsheet
Send the Stream Summary to the flowsheet.

Material Heat Load Work Vol.% Curves Wt.% Curves Petroleum Polymers Solids

	Units	LIQNH3	OFFGAS	S2	S3	S4	S5	S6
Description								
From		FLSH-100	FLSH-100	C-100	M-100	E-100	R-100	E-101
To				M-100	E-100	R-100	E-101	FLSH-
Stream Class		CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONV
Maximum Relative Error								
Cost Flow	\$/hr							
- MIXED Substream								
Phase		Liquid	Vapor	Vapor	Vapor	Vapor	Vapor	Mixed
Temperature	K	300.15	300.15	1239.98	1239.98	755	755	
Pressure	atm	270	270	271.4	271.4	271.4	270	
Molar Vapor Fraction		0	1	1	1	1	1	
Molar Liquid Fraction		1	0	0	0	0	0	
Molar Solid Fraction		0	0	0	0	0	0	
Mass Vapor Fraction		0	1	1	1	1	1	

Results Available (problem not yet run) Check Status

100%

Type here to search

EN 4:40 PM 6/28/2020

Stream summary
exported to the
flowsheet in table
form

Example5.1.bkp - Aspen Plus V9 - aspenONE

File Home Economics Dynamics Equation Oriented View Customize Resources Modify Format

Rotate Flip Horizontal Flip Vertical Reconnect Break Insert Join Reroute Stream Align 3D Icons Temperature Heat/Work Show Status Unit Operations Mass Flow Rate Pressure Vapor Fraction Stream Results Mole Flow Rate Volume Flow Rate Display Options Lock Flowsheet Section Hierarchy View Parent View Child Import Export Move Selection

Simulation Capital: ___USD Utilities: ___USD/Year Energy Savings: ___MW (___%) Exchangers - Unknown: 0 OK: 0 Risk: 0

Main Flowsheet FLSH-100 (Flash2) - Input R-100 (RStoic) Results Summary - Streams (All)

SYNGAS C-100 S2 M-100 S3 E-100 S4 R-100 S5 E-101 S6 FLSH-100 LIQNH3 OFFGAS

	Units	LIQNH3	OFFGAS	S2	S3	S4	S5	S6	SYNGAS
Description		FLSH-100	FLSH-100	C-100	M-100	E-100	R-100	E-101	FLSH-100
From		CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN
To									
Stream Class		CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN
Maximum Relative Error									
Cost Flow	\$/hr								
MIXED Substream									
Phase		Liquid	Vapor	Vapor	Vapor	Vapor	Vapor	Mixed	Vapor
Temperature	K	300.15	300.15	1239.98	1239.98	755	755	300.15	553.15
Pressure	atm	270	270	271.4	271.4	270	270	270	26.17
Molar Vapor Fraction		0	1	1	1	1	1	0.813785	1
Molar Liquid Fraction		1	0	0	0	0	0	0.186215	0
Molar Solid Fraction		0	0	0	0	0	0	0	0
Mass Vapor Fraction		0	1	1	1	1	1	0.712155	1
Mass Liquid Fraction		1	0	0	0	0	0	0.287845	0
Mass Solid Fraction		0	0	0	0	0	0	0	0
Molar Enthalpy	cal/mol	-15559.8	-1273.62	6734.54	6734.54	3108.23	654.733	-3933.91	1550.27
Mass Enthalpy	cal/gm	-921.763	-133.271	768.889	768.889	354.869	59.955	-360.235	176.996
Molar Entropy	cal/mol-K	-44.4177	-11.7702	0.250533	0.250533	-3.44914	-7.8257	-17.8496	-1.01965
Mass Entropy	cal/gm-K	-2.63131	-1.23163	0.0286036	0.0286036	-0.393792	-0.716612	-1.63452	-0.116437
Molar Density	mol/cc	0.0286241	0.00949913	0.0025293	0.0025293	0.00402264	0.00401231	0.0108426	0.000570106
Mass Density	gm/cc	0.483187	0.0907221	0.0221537	0.0221537	0.0352335	0.0438161	0.118405	0.00499344
Enthalpy Flow	cal/sec	-4.51875e+06	-1.6164e+06	1.30949e+07	1.30949e+07	6.04378e+06	1.02109e+06	-6.13515e+06	3.01441e+06
Average MW		16.8804	9.5566	8.75879	8.75879	8.75879	10.9204	10.9204	8.75879
Mole Flows	kmol/hr	1045.49	4568.91	7000	7000	7000	5614.4	5614.4	7000
Mole Fractions									

Results Available (problem not yet run) Check Status

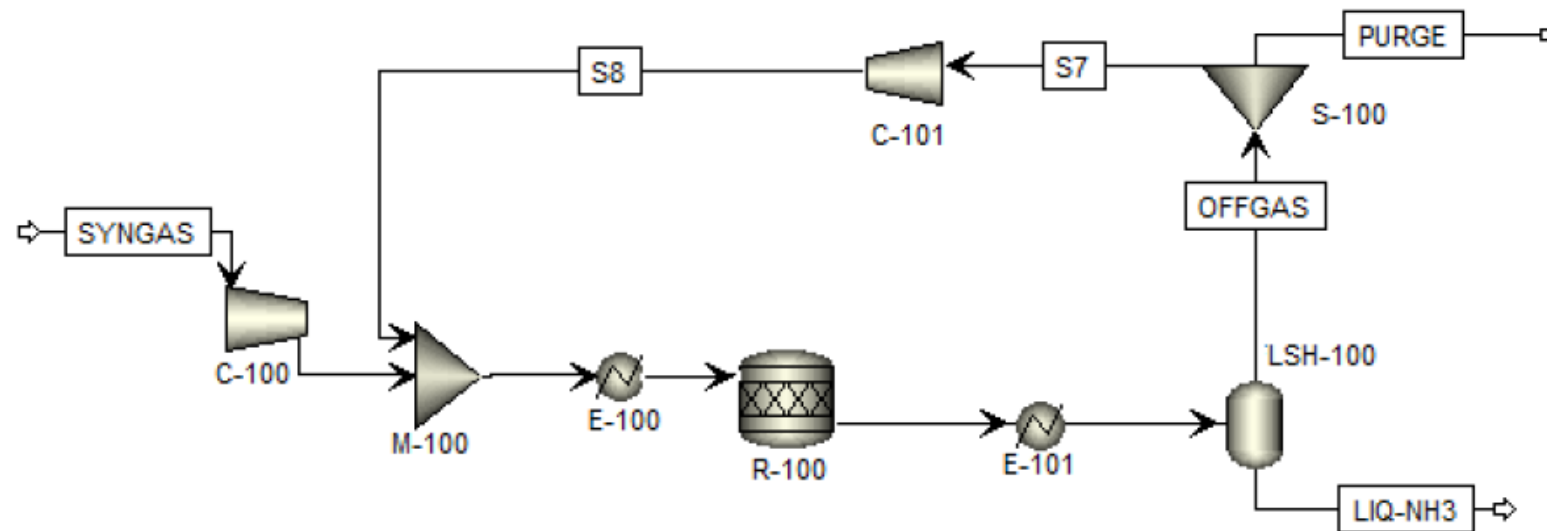
67%

Type here to search

EN 4:43 PM 6/28/2020

Example 4.3 (Process with recycle)

- This process produces two outlet streams; a liquid stream containing the ammonia product and a vapor stream containing mostly unreacted hydrogen and nitrogen. It is desired to capture and recycle these unreacted materials to minimize costs and maximize product yield.



Splitter S-100

- Enter a value of 0.01 for the Split fraction of the purge stream. This means that 1% of the OFFGAS stream will be diverged to the purge stream.

Compressor C-101

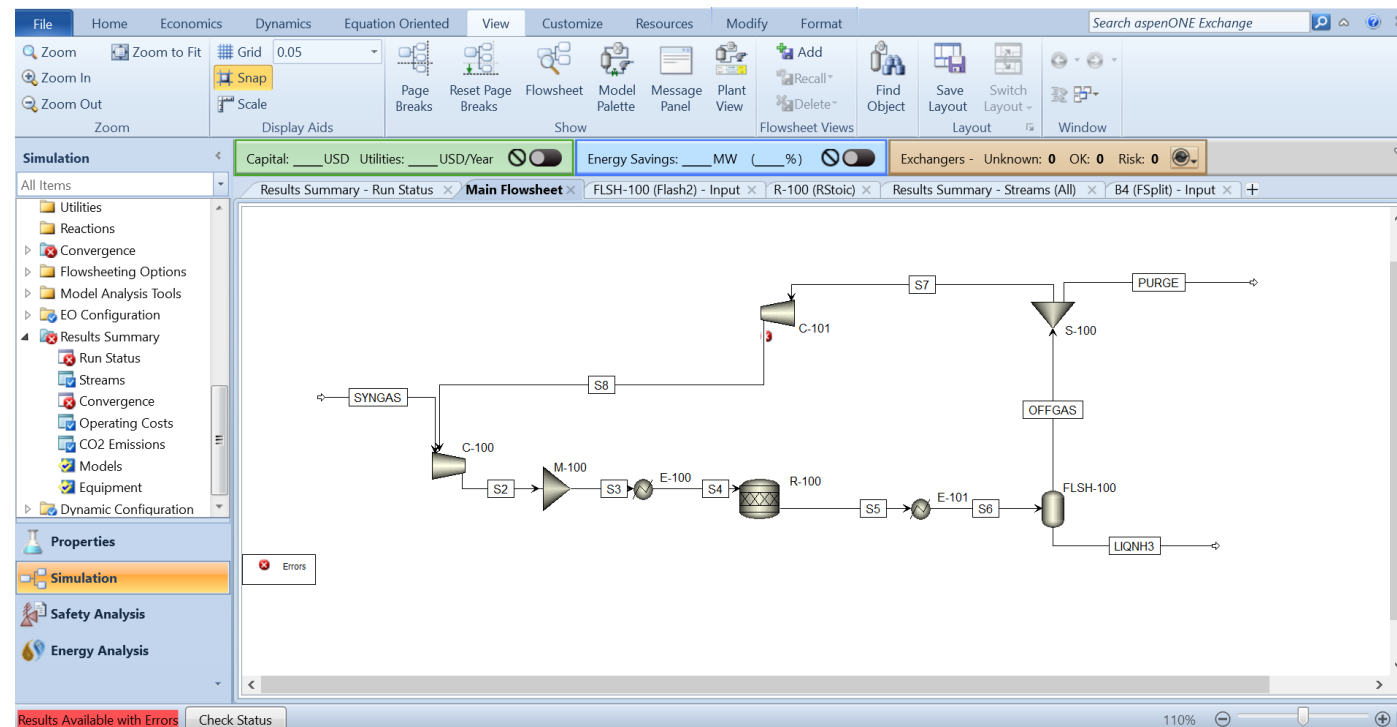
- Select Isentropic as the Type and enter a Discharge pressure of 271 atmospheres.

Problem Analysis

- The simulation could build up from the simulation file of Example 4.3 by adding splitter and compressor
- Equipment model
 - Splitter S-100: FSplit
 - Compressor C-101: Compressor

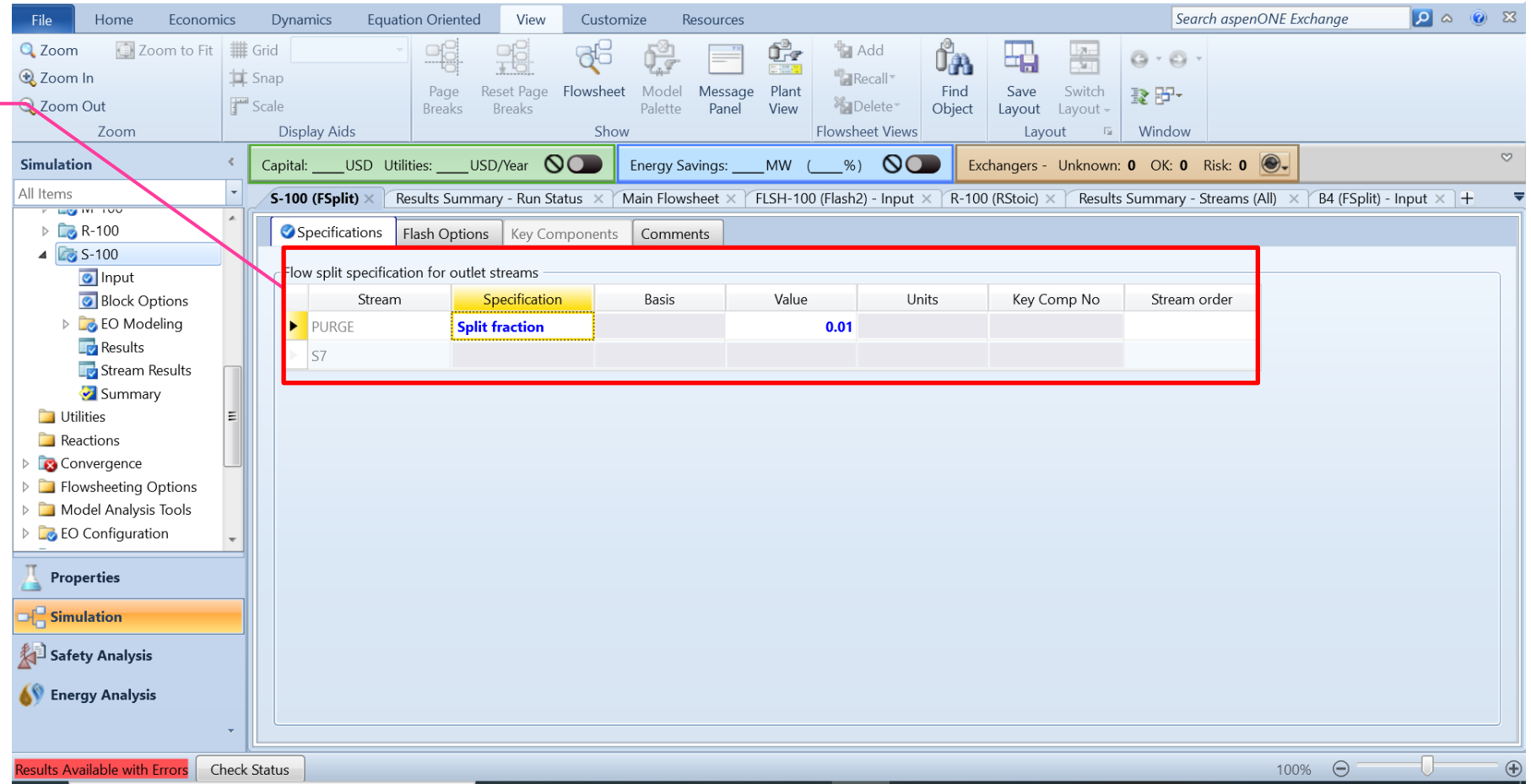
Problem Solving

- Open the simulation file of Example 4.3
- **Insert new equipment and streams into the flow sheet** > include block spec > run simulation



- Insert new equipment and streams into the flow sheet> include block spec> run simulation

Insert the splitter specification based on the info provided in the question



Simulation

Capital: ___USD Utilities: ___USD/Year Energy Savings: ___MW (___%) Exchangers - Unknown: 0 OK: 0 Risk: 0

S-100 (FSplit) Results Summary - Run Status Main Flowsheet FLSH-100 (Flash2) - Input R-100 (RStoic) Results Summary - Streams (All) B4 (FSplit) - Input

Specifications Flash Options Key Components Comments

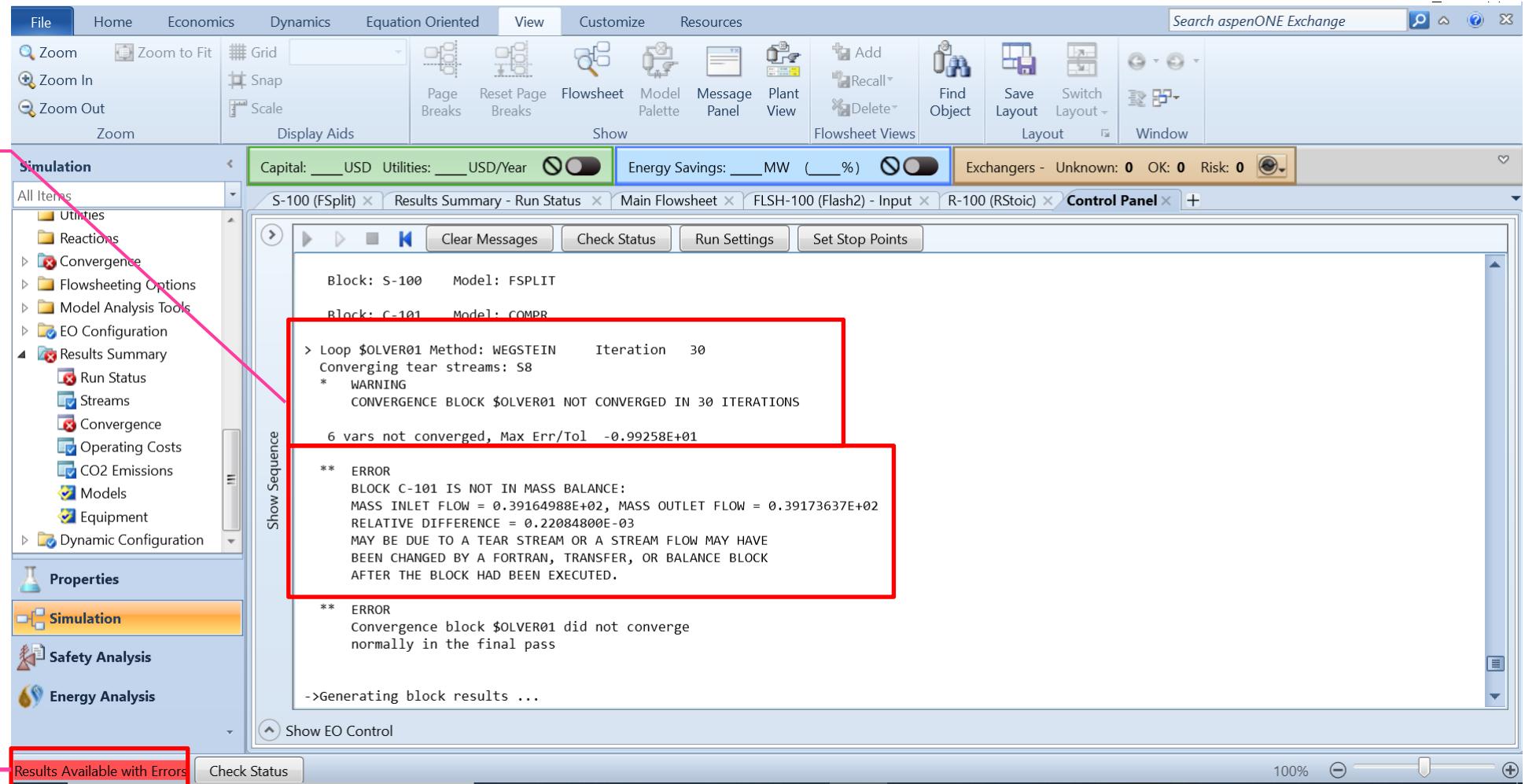
Flow split specification for outlet streams

Stream	Specification	Basis	Value	Units	Key Comp No	Stream order
PURGE	Split fraction		0.01			
S7						

Results Available with Errors Check Status 100%

- Insert new equipment and streams into the flow sheet> include block spec> run simulation

Result available with errors because the solver cannot converge.



The screenshot displays the AspenONE software interface. The left sidebar shows the 'Simulation' tab selected, with 'Results Summary' expanded. The main window shows the 'Control Panel' for the simulation. The 'Run Status' section indicates that the simulation has failed to converge. The error messages are as follows:

```

Block: S-100 Model: FSPLIT
Block: C-101 Model: COMPB
> Loop $SOLVER01 Method: WEGSTEIN Iteration 30
Converging tear streams: S8
* WARNING
CONVERGENCE BLOCK $SOLVER01 NOT CONVERGED IN 30 ITERATIONS
6 vars not converged, Max Err/Tol -0.99258E+01

** ERROR
BLOCK C-101 IS NOT IN MASS BALANCE:
MASS INLET FLOW = 0.39164988E+02, MASS OUTLET FLOW = 0.39173637E+02
RELATIVE DIFFERENCE = 0.22084800E-03
MAY BE DUE TO A TEAR STREAM OR A STREAM FLOW MAY HAVE
BEEN CHANGED BY A FORTRAN, TRANSFER, OR BALANCE BLOCK
AFTER THE BLOCK HAD BEEN EXECUTED.

** ERROR
Convergence block $SOLVER01 did not converge
normally in the final pass

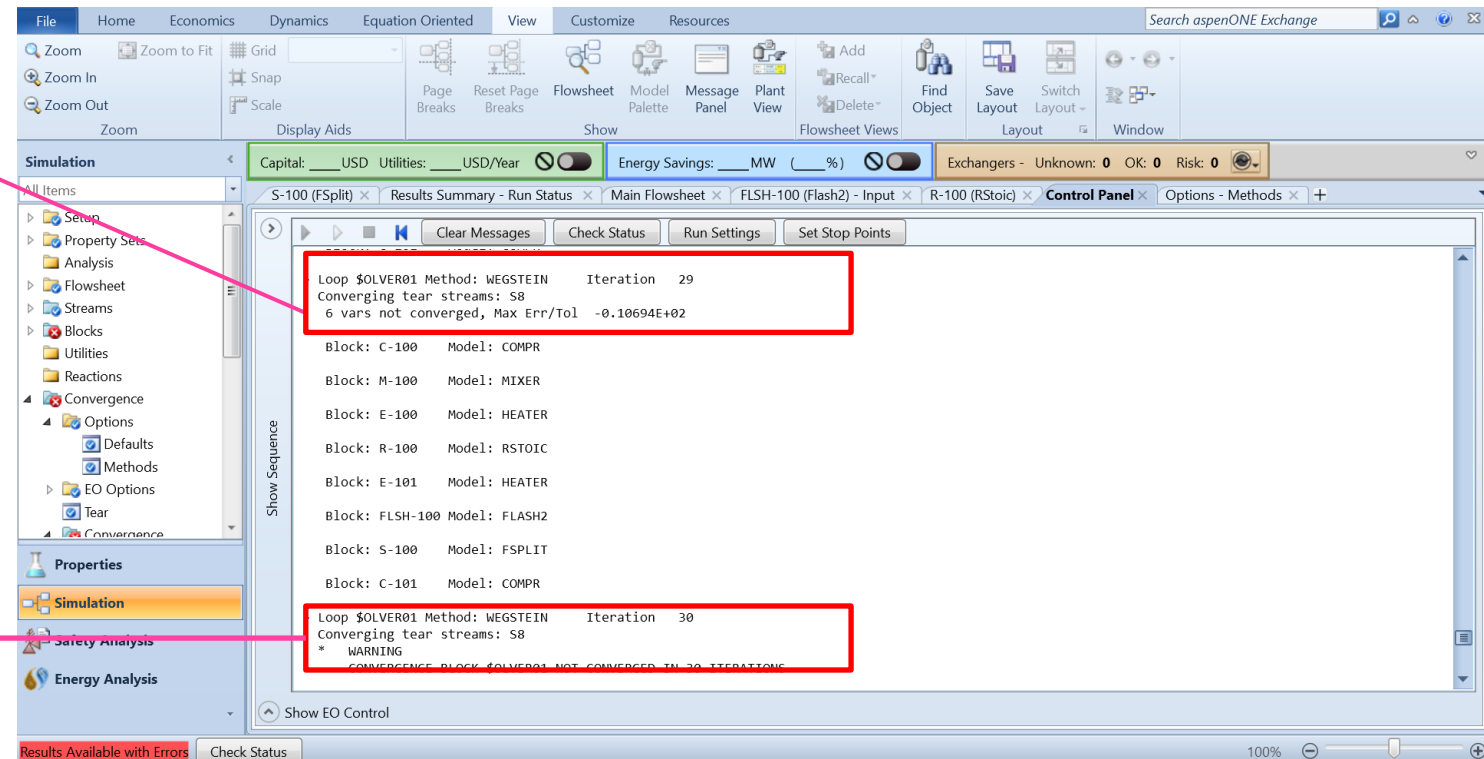
->Generating block results ...
  
```

A red box highlights the 'Results Available with Errors' status in the bottom left corner of the interface.

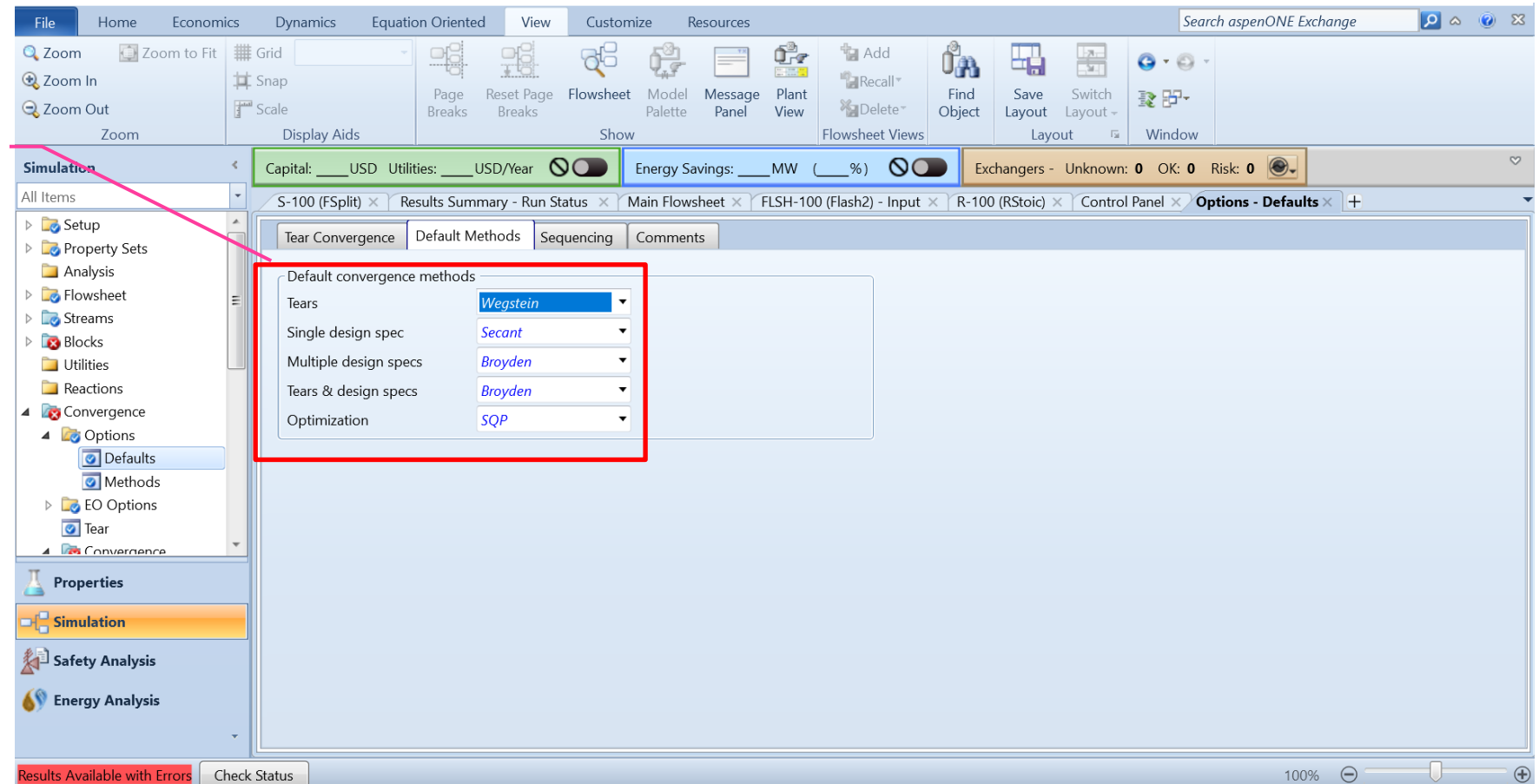
Problem Troubleshooting

- The error due to non-convergence may be attributed to the following reasons:
 - Inappropriate numerical method for convergence
 - Insufficient number of iterations

Wegstein method is a good method to use when trying to converge a single recycle stream, and stream S8 is an appropriate stream to attempt to converge.

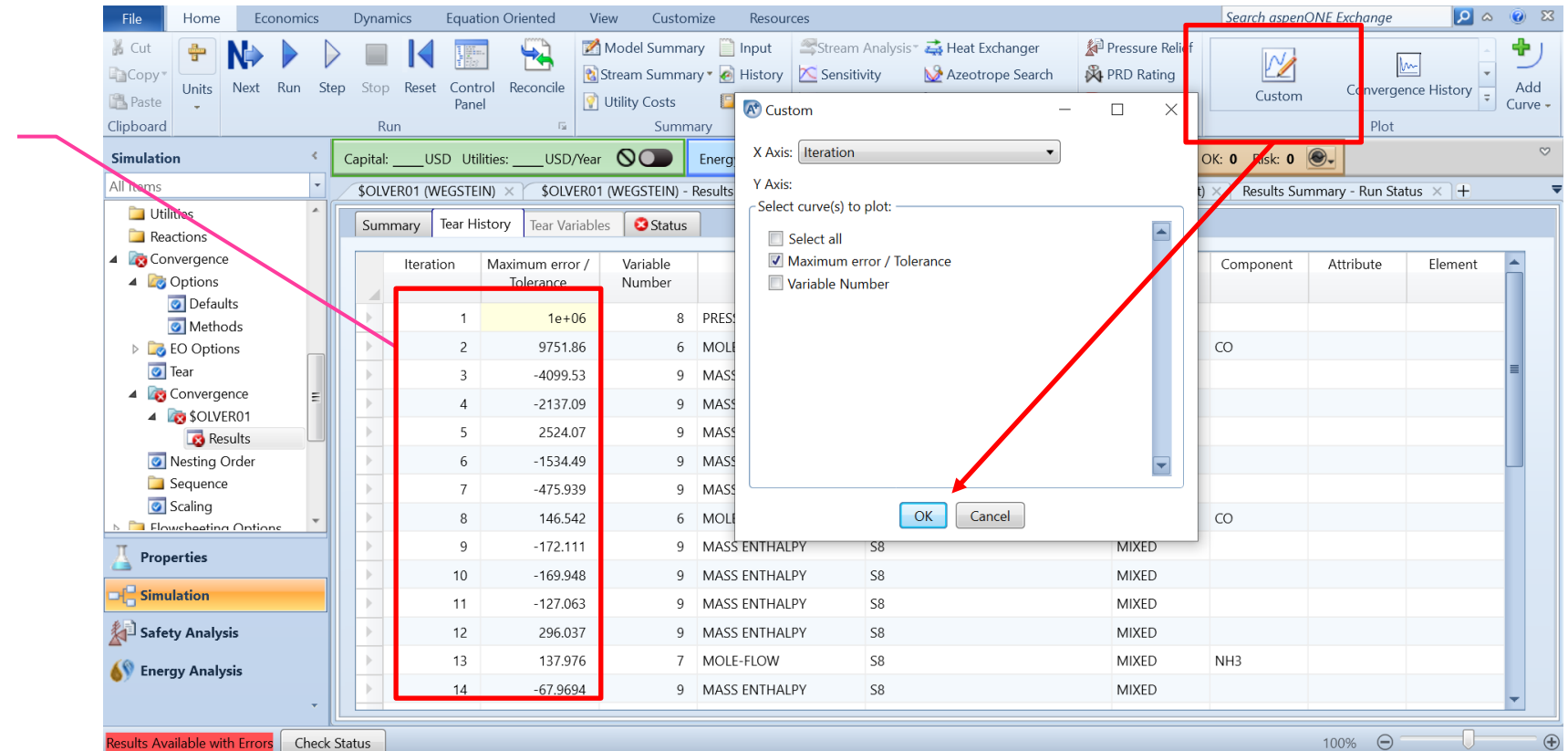


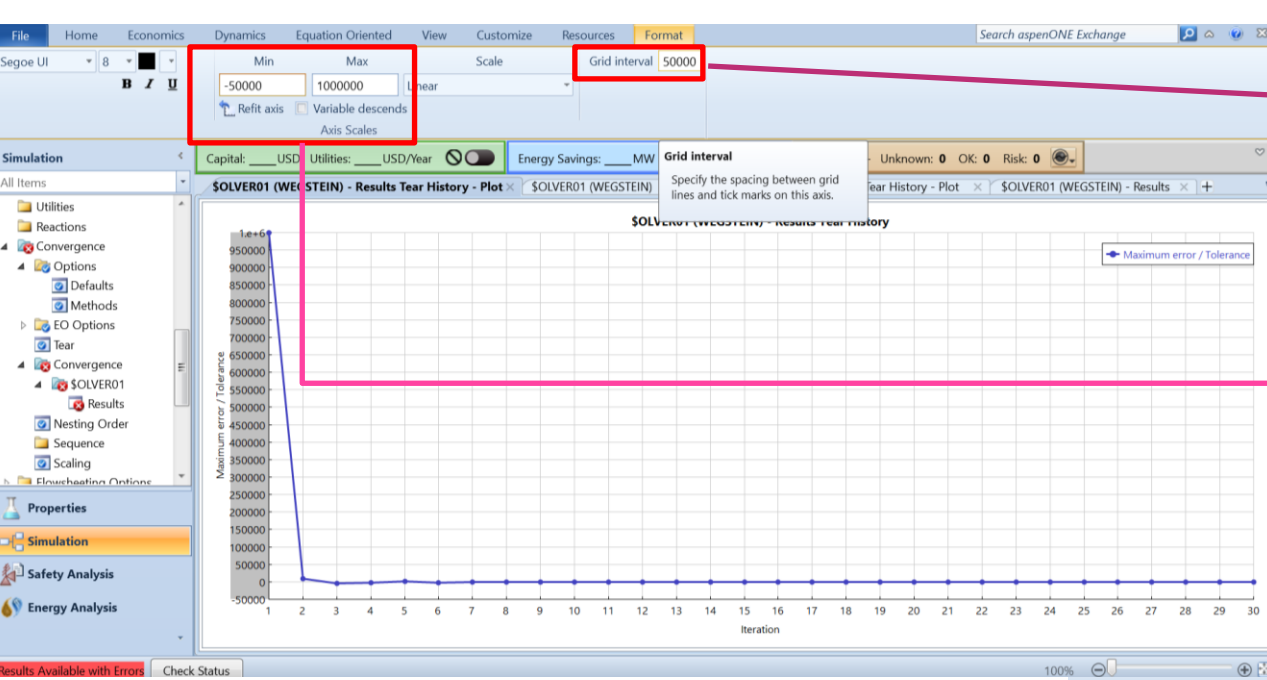
The method used can also be checked in Convergence>Options>Default. Wegstein is the default convergence method for system generated tear convergence blocks and is usually the quickest and most reliable method for tear stream convergence.



- The error due to non-convergence may be attributed to the following reasons:
 - Inappropriate numerical method for convergence
 - **Insufficient number of iterations**

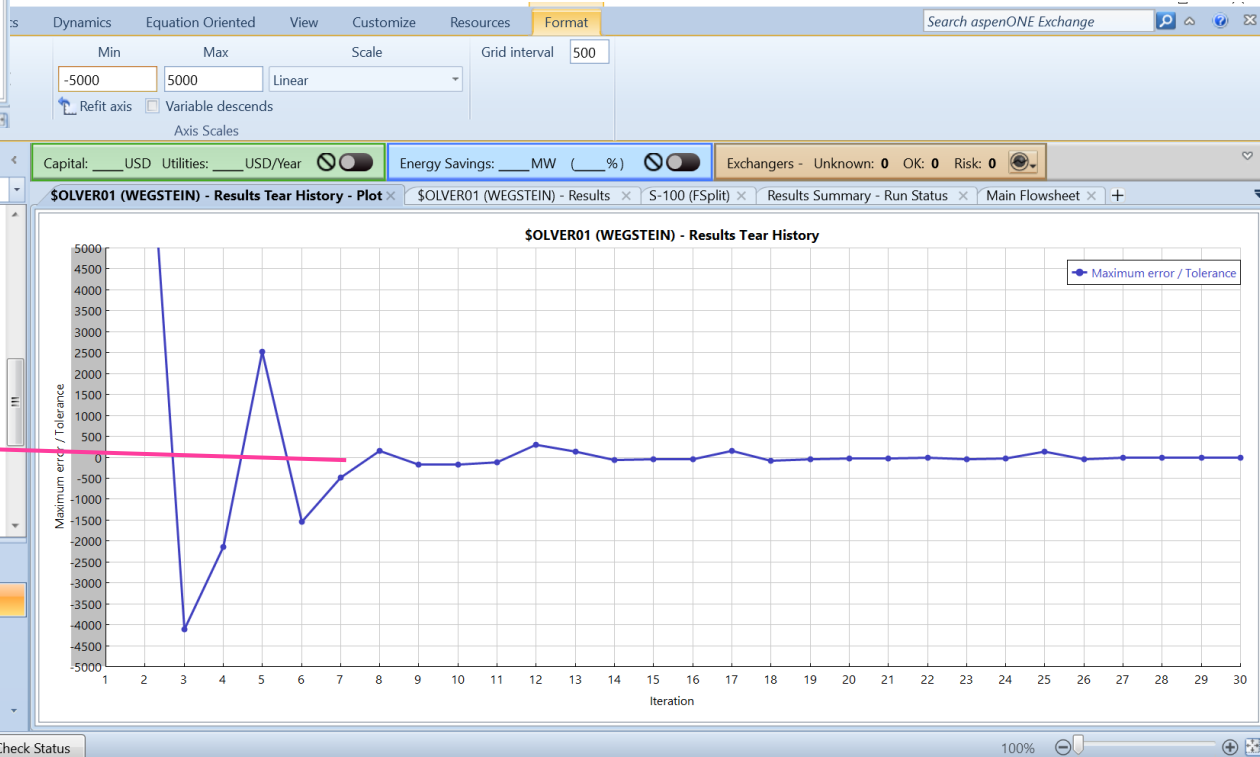
To validate whether the non-convergence is due to the no. of iteration, plot the maximum error/tolerance vs no. of iteration obtained from Convergence>Convergence >Results.





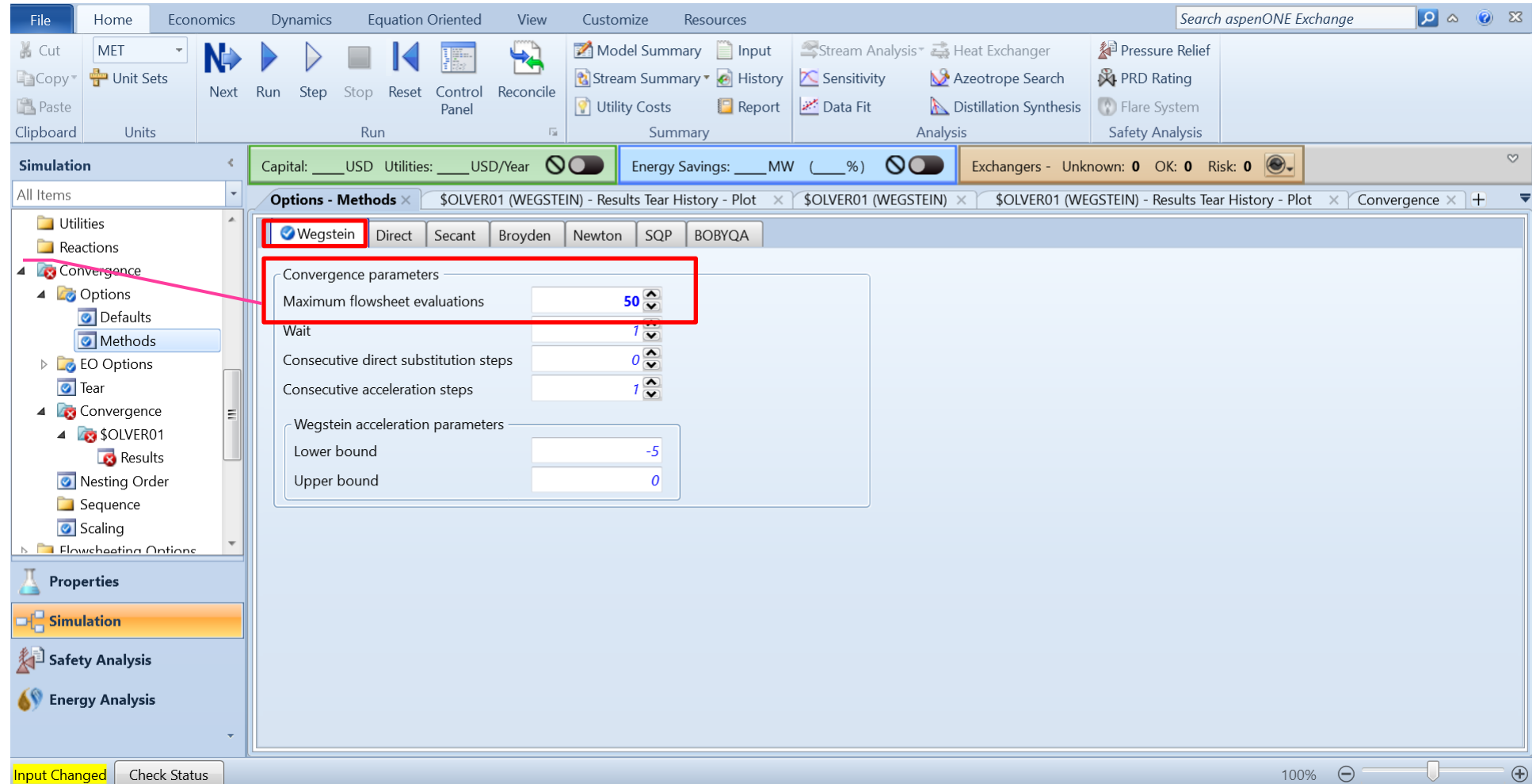
For a more meaningful graphical representation, the change the y-axis for its:

1. Grid interval to 500
2. Min.: -5000, Max.: 5000



The reducing trend of the max. error/tolerance (towards 0) with the number of iteration indicates that the non-convergence can be solved by increasing the number of iteration.

In the Convergence
 >Options > Methods
 >Wegstein, change the
 max. flowsheet
 evaluation from 30 to 50



The screenshot displays the AspenONE software interface. The 'Options - Methods' window is open, showing the 'Wegstein' method selected. The 'Convergence parameters' section is highlighted with a red box, showing the 'Maximum flowsheet evaluations' set to 50. Other parameters include 'Wait' (1), 'Consecutive direct substitution steps' (0), 'Consecutive acceleration steps' (1), 'Wegstein acceleration parameters' (Lower bound: -5, Upper bound: 0). The left sidebar shows the 'Convergence' folder expanded, with '\$OLVER01' selected. The top menu bar includes File, Home, Economics, Dynamics, Equation Oriented, View, Customize, and Resources. The bottom status bar indicates 'Input Changed' and 'Check Status'.

Reset and run the simulation again and the results are available without errors now.

The screenshot displays the Aspen Plus software interface. The top menu bar includes File, Home, Economics, Dynamics, Equation Oriented, View, Customize, and Resources. The ribbon contains various toolbars for simulation control and analysis. The left sidebar shows the 'Simulation' section with a tree view of items including Utilities, Reactions, Convergence, Options, EO Options, and Convergence. The 'Properties' section is also visible. The main window shows the 'Control Panel' with tabs for Options - Methods, S-100 (FSplit), Results Summary - Run Status, Main Flowsheet, FLSH-100 (Flash2) - Input, and R-100 (RStoic). The 'Run Status' tab is active, displaying the following text:

```

Block: S-100 Model: FSPLIT

Block: C-101 Model: COMPR

> Loop $OLVER01 Method: WEGSTEIN Iteration 49
Converging tear streams: S8
# Converged Max Err/Tol -0.54904E+00

->Generating block results ...

Block: E-100 Model: HEATER

Block: E-101 Model: HEATER

->Simulation calculations completed ...

*** No Warnings were issued during Input Translation ***

*** No Errors or Warnings were issued during Simulation ***

->Generating results ...
  
```

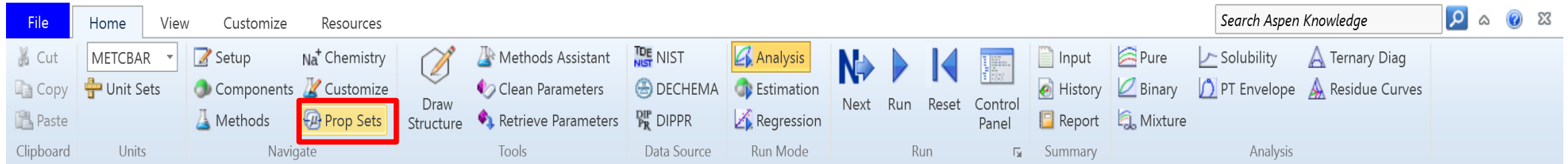
At the bottom left, the 'Results Available' button is highlighted with a red box, and a yellow line connects it to the text box on the left. The 'Check Status' button is also visible next to it.

4.3 Property Sets

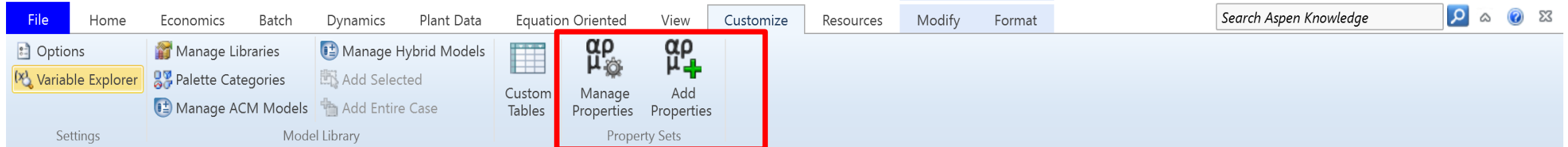
Property Sets

- A property set (Prop-Set) is a way of accessing a collection of properties as an object with a user-given name
- Only the name of the property set is referenced, when using the properties in an application
- Use property sets to report thermodynamics, transport, and other property values
- Prop-Set are available in both Properties and Simulation Environments
 - Properties Environment on the Home ribbon Navigate group
 - Simulation Environment in the Property Sets folder in the Navigation Pane
- Prop-Sets are the same in both environment

Property Sets



or

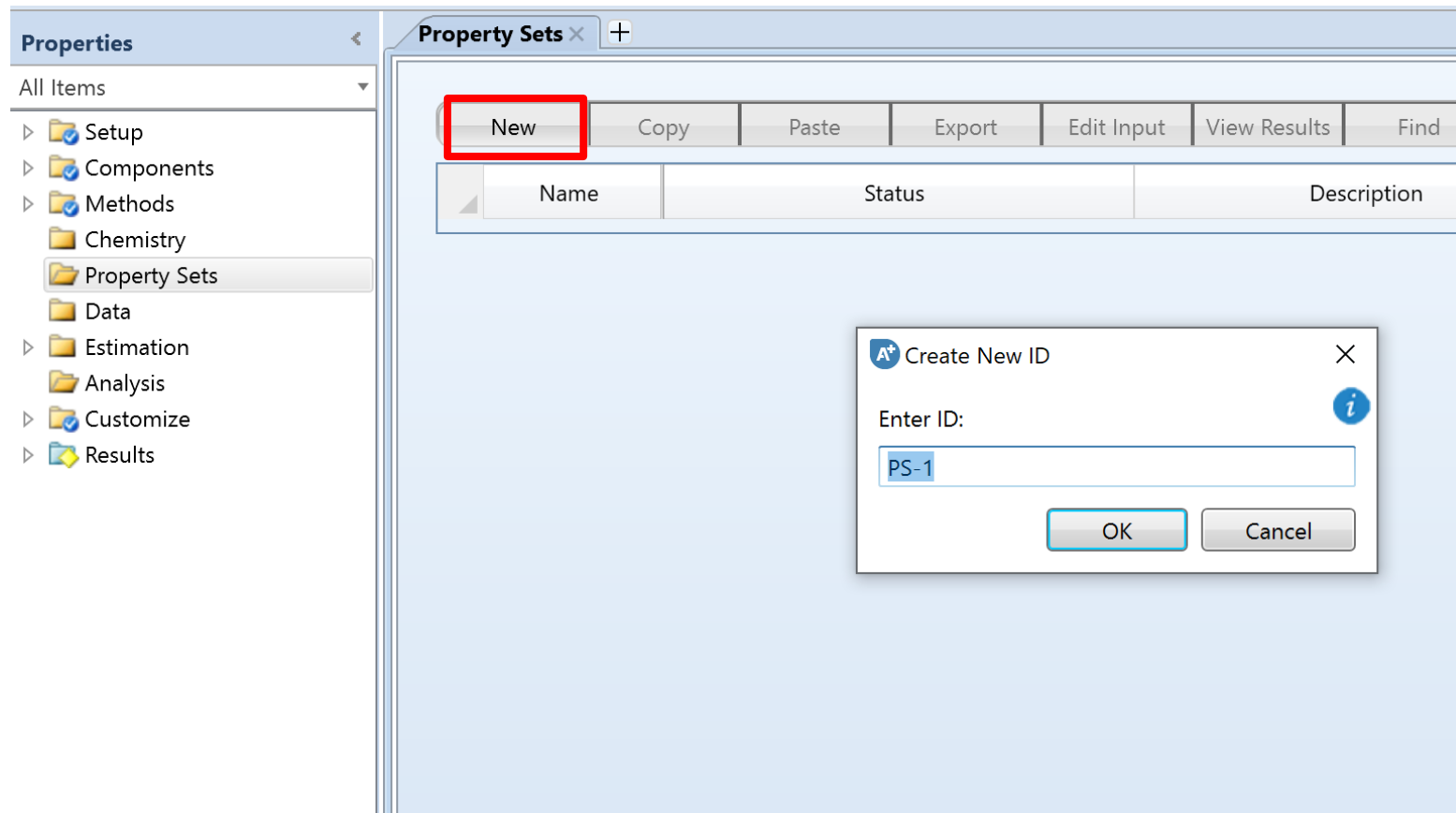


Properties Included in Prop-Sets

- Available properties include:
 - Thermodynamic properties of components in a mixture
 - Pure components thermodynamics properties
 - Transport properties
 - Electrolyte properties
 - Petroleum-related properties
- Properties commonly included in property sets include:
 - Molar vapor fraction of a stream (VFRAC)
 - Fraction of Liquid1 to total liquid for mixture (BETA)
 - Constant pressure heat capacity for a mixture (CPMX)
 - Viscosity of a mixture (MUMX)

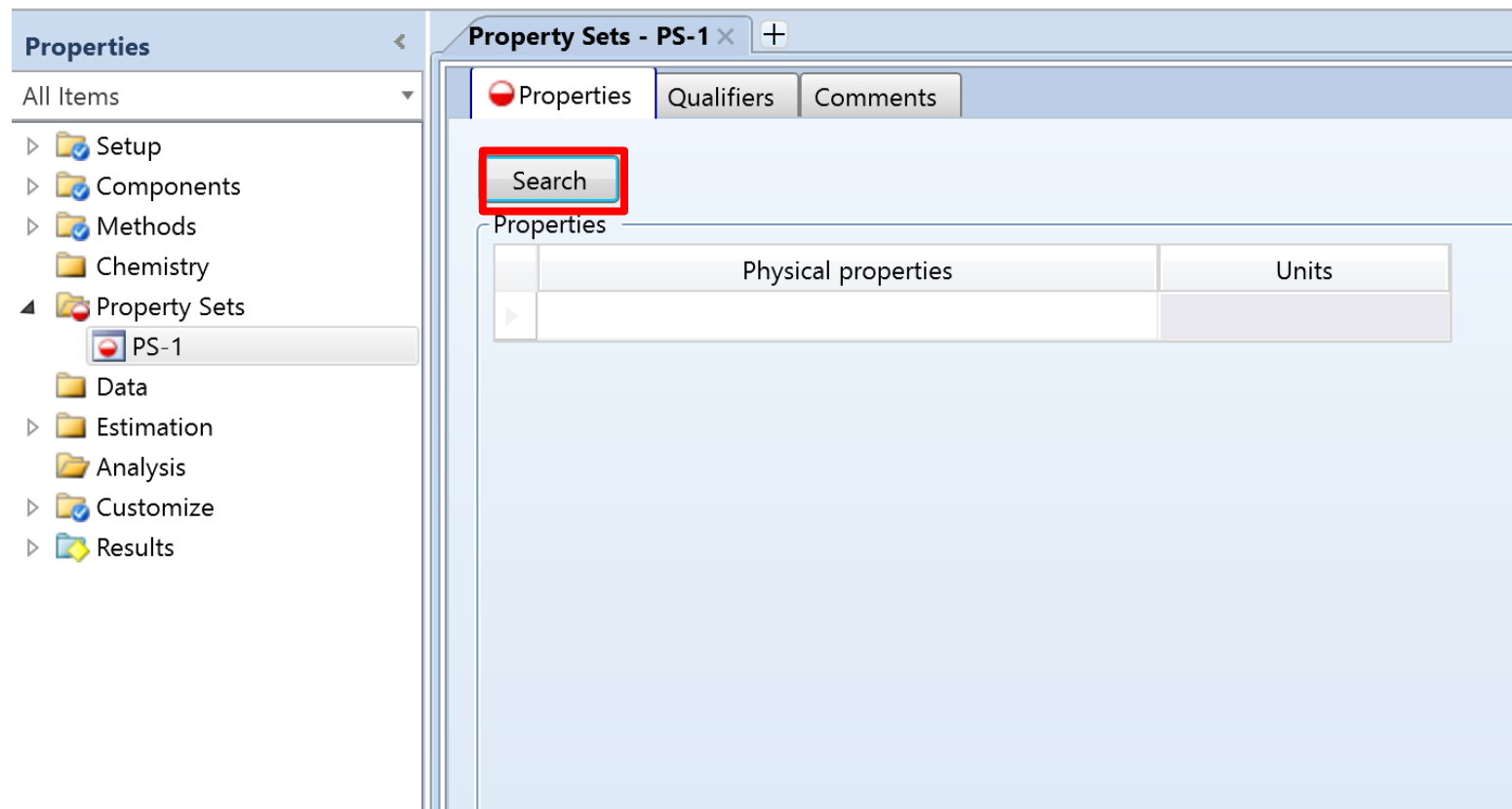
Specifying Property Sets

- Create new property set using **Property Sets | New** and enter the ID name



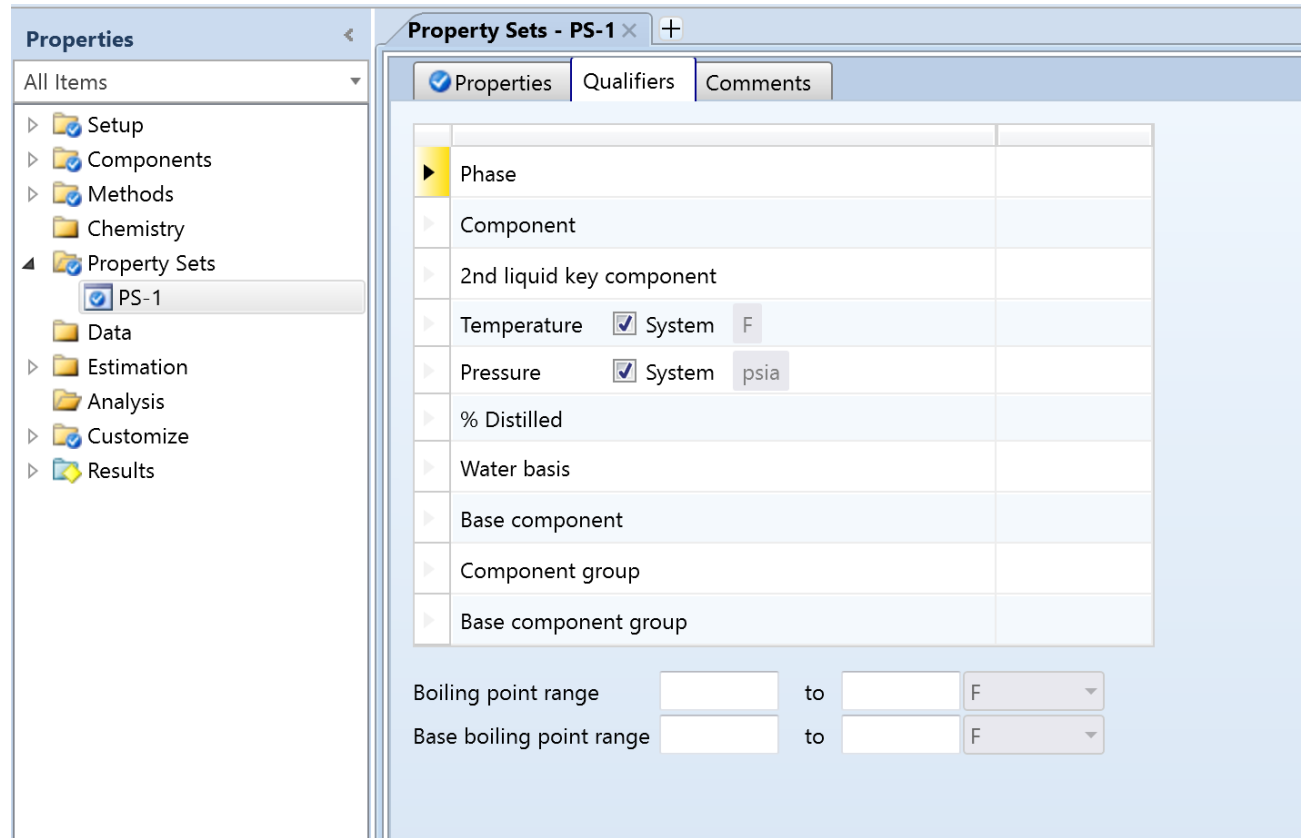
Specifying Property Sets

- Use the search button to search for a property



Specifying Property Sets

- All specified qualifiers apply to each selected property, where applicable



Property Sets - PS-1

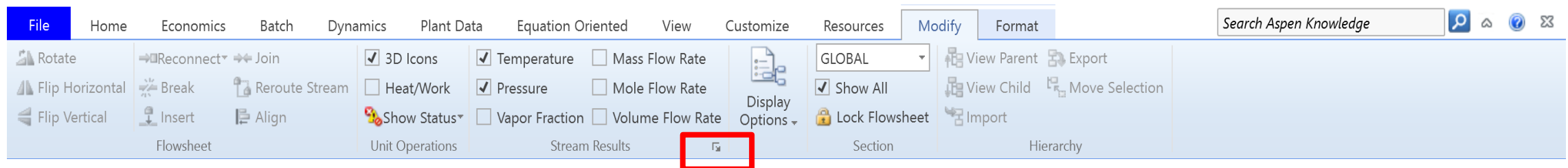
Properties	Qualifiers	Comments
Phase		
Component		
2nd liquid key component		
Temperature	☒ System F	
Pressure	☒ System psia	
% Distilled		
Water basis		
Base component		
Component group		
Base component group		

Boiling point range to F

Base boiling point range to F

Additional Global Variables

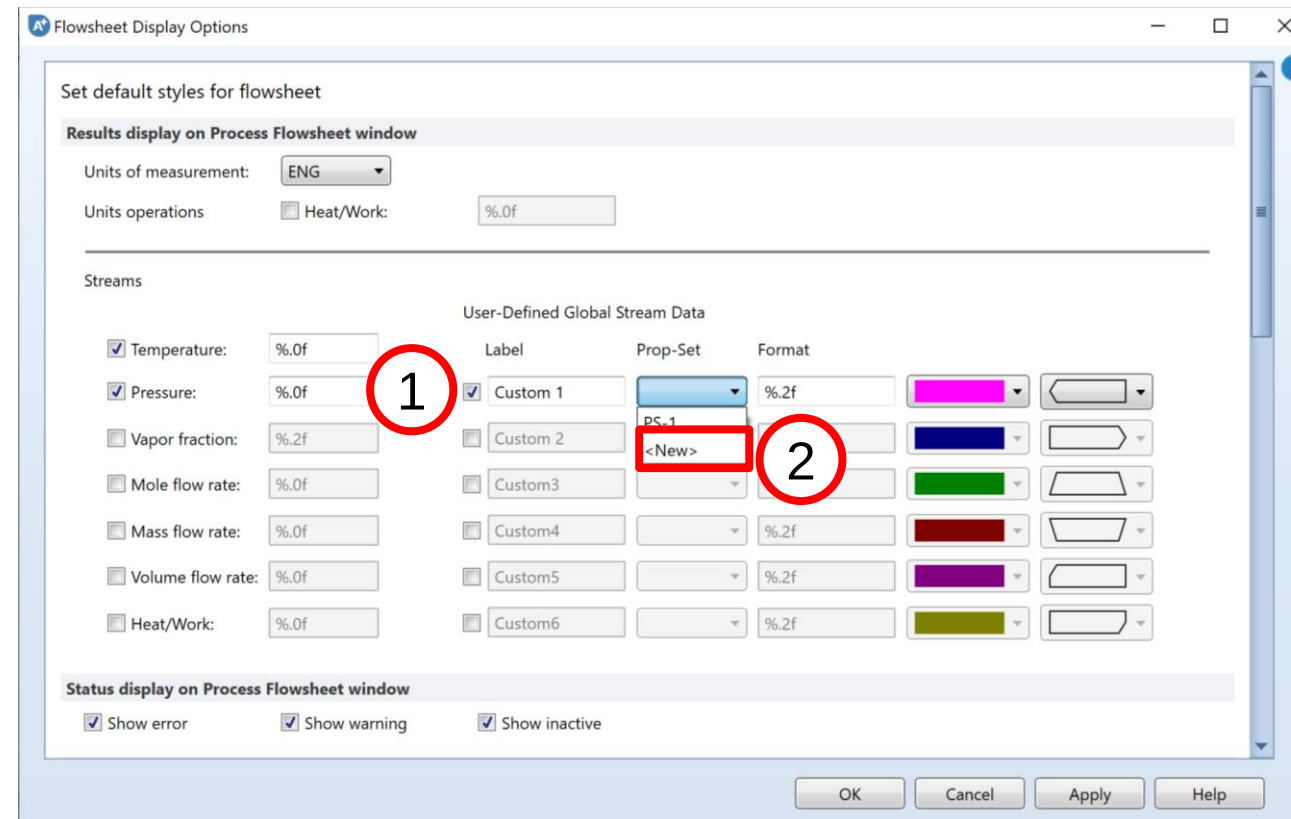
- Global variables can be added to the Aspen Plus flowsheet (temperature, pressure, flow rates, vapor fraction, etc.)
- Up to six additional variables, via Property Sets can be included with regularly available variables
- Add the variables from the **Stream Results** group of the **Flowsheet | Modify** ribbon



- Open the **File | Options | Flowsheet Display Options** form

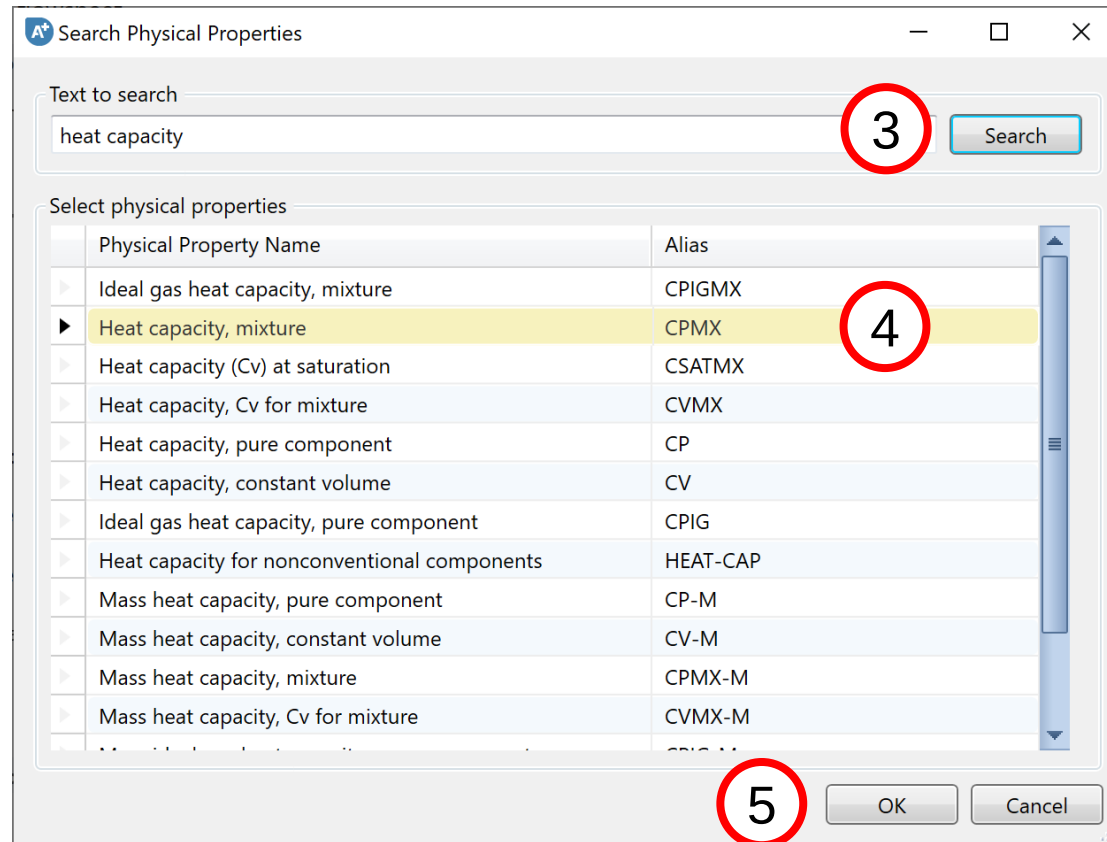
Additional Global Variables

- Define a Prop-Set with a single variable from **File | Options | Flowsheet Display Options**



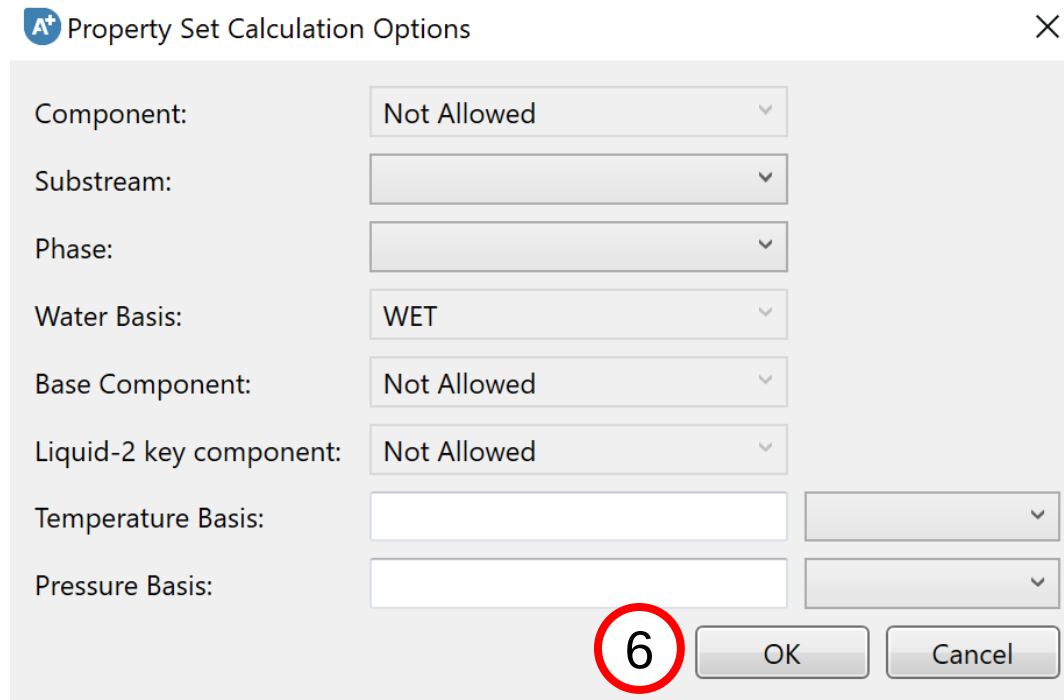
Additional Global Variables

- Search property to display and select



Additional Global Variables

- Select any preferred units or qualifiers
 - If no units are selected, system default units of measure will be used
 - If no qualifiers are set, property defaults will be calculated



Property Set Calculation Options

Component: Not Allowed

Substream:

Phase:

Water Basis: WET

Base Component: Not Allowed

Liquid-2 key component: Not Allowed

Temperature Basis:

Pressure Basis:

6 OK Cancel

Additional Global Variables

- Customize display options for Global Data variable

Flowsheet Display Options

Set default styles for flowsheet

Results display on Process Flowsheet window

Units of measurement: ENG

Units operations ☐ Heat/Work: %.0f

Streams

☐ Temperature: %.0f

☐ Pressure: %.0f

☐ Vapor fraction: %.2f

☐ Mole flow rate: %.0f

☐ Mass flow rate: %.0f

☐ Volume flow rate: %.0f

☐ Heat/Work: %.0f

User-Defined Global Stream Data

Label	Prop-Set	Format	Color	Shape
☒ Heat Capacity	PSGD-1	%.2f	Magenta	Circle
☐ Custom2		%.2f	Blue	Circle
☐ Custom3		%.2f	Green	Circle
☐ Custom4		%.2f	Red	Circle
☐ Custom5		%.2f	Purple	Circle
☐ Custom6		%.2f	Olive	Circle

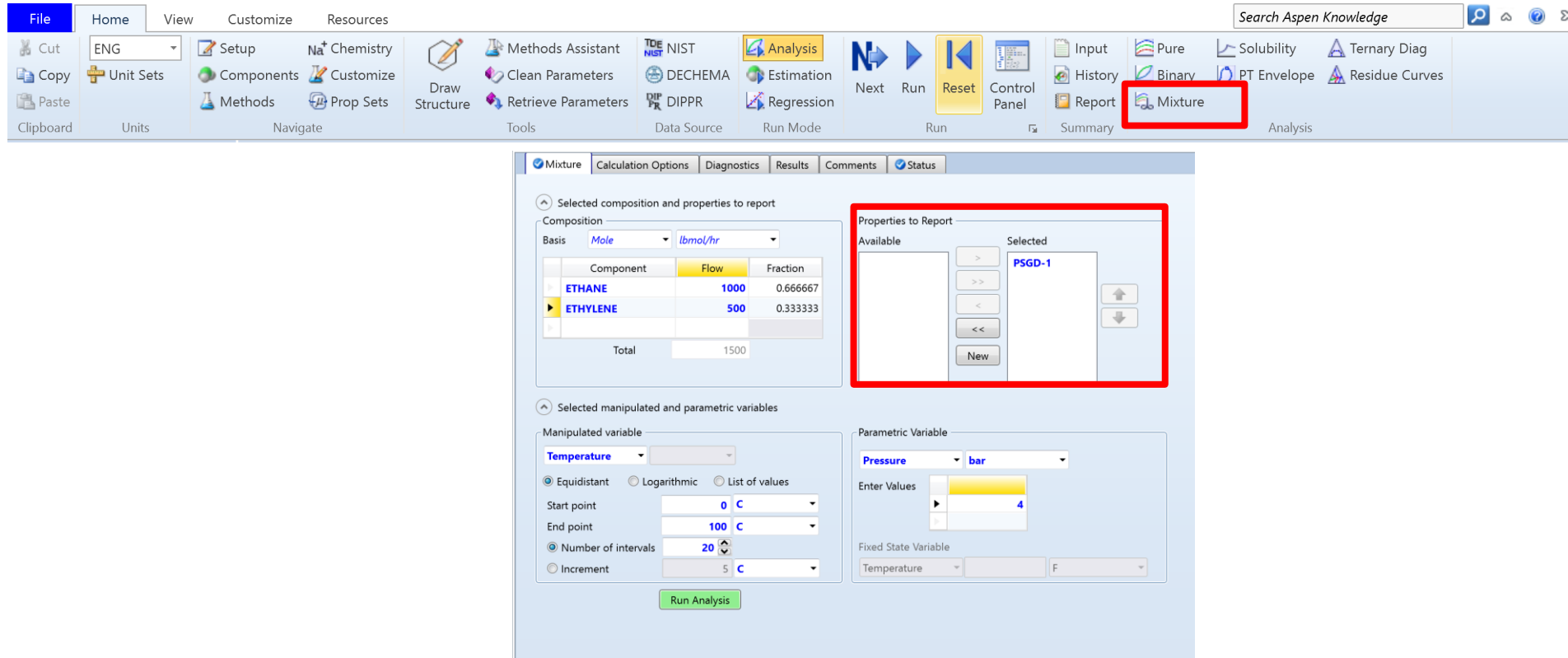
Status display on Process Flowsheet window

☒ Show error ☒ Show warning ☒ Show inactive

OK Cancel Apply Help

Properties for Mixture Analysis

- Interactive Mixture Analysis requires the user to create a Prop-Set for the properties reported



The screenshot shows the Aspen Plus software interface. The top menu bar includes File, Home, View, Customize, and Resources. The ribbon contains various toolbars: Clipboard, Units, Navigate, Tools, Data Source, Run Mode, Run, Summary, and Analysis. The Analysis toolbar is highlighted, and the 'Mixture' button is circled in red.

The main window displays the 'Mixture' tab with the following settings:

- Selected composition and properties to report:**
 - Basis: Mole
 - Units: lbmol/hr
 - Component Flow Fraction:

Component	Flow	Fraction
ETHANE	1000	0.666667
ETHYLENE	500	0.333333
Total	1500	
- Properties to Report:**
 - Available: (Empty list)
 - Selected: PSGD-1
- Selected manipulated and parametric variables:**
 - Manipulated variable: Temperature
 - Equidistant: ☒ (Selected)
 - Logarithmic: ☐
 - List of values: ☐
 - Start point: 0 C
 - End point: 100 C
 - Number of intervals: 20
 - Increment: 5 C
 - Parametric Variable: Pressure
 - bar
 - Enter Values: 4
 - Fixed State Variable: Temperature

A 'Run Analysis' button is located at the bottom of the main window.

Exercise 4.2